

# Projections and Linear Systems

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## Design Matrix

**Definition 0.1** (Design matrix (model matrix)). In a regression model with  $n$  observations and  $p$  predictors, the **design matrix** (or **model matrix**)  $\mathbf{X}$  is the  $n \times p$  matrix whose  $i$ -th row is the covariate vector  $\tilde{x}_i^\top$  for observation  $i$ :

$$\begin{aligned}\mathbf{X} &= \begin{bmatrix} \tilde{x}_1^\top \\ \tilde{x}_2^\top \\ \vdots \\ \tilde{x}_n^\top \end{bmatrix} \\ &= \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}\end{aligned}$$

*Remark 0.1* (Products with the design matrix). The product  $\mathbf{X}\tilde{\beta}$  collects the values  $\tilde{x}_i^\top \tilde{\beta}$  for all  $n$  observations into a single  $n \times 1$  vector:

$$\mathbf{X}\tilde{\beta} = \begin{bmatrix} \tilde{x}_1^\top \tilde{\beta} \\ \vdots \\ \tilde{x}_n^\top \tilde{\beta} \end{bmatrix}$$

For example, take the rank-2 matrix from [Example 2 in Subspaces and Rank](#) as  $\mathbf{X}$ , with  $n = 3$  observations and  $p = 2$  columns: an intercept column and one covariate. With  $\tilde{\beta} = (1, 2)$ ,

$$\begin{aligned}\mathbf{X}\tilde{\beta} &= \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}.\end{aligned}$$

The matrix  $\mathbf{X}^\top \mathbf{X}$  is  $p \times p$  and symmetric, since

$$\begin{aligned}(\mathbf{X}^\top \mathbf{X})^\top &= \mathbf{X}^\top (\mathbf{X}^\top)^\top \\ &= \mathbf{X}^\top \mathbf{X}\end{aligned}$$

([Theorem 16 in Matrices](#)). When  $\mathbf{X}^\top \mathbf{X}$  is invertible, it appears in the OLS estimator  $\hat{\tilde{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \tilde{y}$ .

**Theorem 0.1** ( $\mathbf{X}^\top \mathbf{X}$  is invertible when  $\mathbf{X}$  has full column rank). *If  $\mathbf{X}$  is an  $n \times p$  matrix with  $\text{rank}(\mathbf{X}) = p$  ([Definition 2 in Subspaces and Rank](#)), then the  $p \times p$  matrix  $\mathbf{X}^\top \mathbf{X}$  is invertible.*

#### Proof

*Proof.* Let  $\tilde{c}$  be a vector of length  $p$  with  $\mathbf{X}^\top \mathbf{X}\tilde{c} = \tilde{0}$ . Then

$$\begin{aligned}0 &= \tilde{c}^\top \mathbf{X}^\top \mathbf{X}\tilde{c} && \text{(multiply } \mathbf{X}^\top \mathbf{X}\tilde{c} = \tilde{0} \text{ on the left by } \tilde{c}^\top) \\ &= (\mathbf{X}\tilde{c})^\top (\mathbf{X}\tilde{c}) && \text{(transpose of a product)} \\ &= \|\mathbf{X}\tilde{c}\|^2 && \text{(definition of the Euclidean norm)}\end{aligned}$$

so  $\mathbf{X}\tilde{c} = \tilde{0}$ . Because  $\mathbf{X}\tilde{c} = c_1(\text{column } 1) + \dots + c_p(\text{column } p)$  and the columns of  $\mathbf{X}$  are linearly independent,  $\tilde{c} = \tilde{0}$ . So the only solution of  $\mathbf{X}^\top \mathbf{X}\tilde{c} = \tilde{0}$  is  $\tilde{c} = \tilde{0}$ , which says that the columns of the square matrix  $\mathbf{X}^\top \mathbf{X}$  are linearly independent, and a square matrix with linearly independent columns is invertible (Banerjee and Roy 2014, Corollary 5.7, p. 143).  $\square$

**Example 0.1** (Inverting  $\mathbf{X}^\top \mathbf{X}$ ). For the rank-2 matrix  $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$  from [Example 2 in Subspaces and Rank](#):

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \quad (\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix},$$

and multiplying the two out gives  $\mathbf{I}_2$ .

**Definition 0.2** (Hat matrix). For an  $n \times p$  design matrix  $\mathbf{X}$  (Definition 0.1) with  $\text{rank}(\mathbf{X}) = p$ , the **hat matrix** is the  $n \times n$  matrix

$$\underset{n \times n}{\mathbf{H}} \stackrel{\text{def}}{=} \underset{n \times p}{\mathbf{X}} \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1}}_{p \times p} \underbrace{\mathbf{X}^\top}_{p \times n}$$

The inverse exists by Theorem 0.1.

**Example 0.2** (The hat matrix of an intercept-only model). With  $n = 2$  observations and only an intercept,  $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  ( $2 \times 1$ , rank 1), so  $\mathbf{X}^\top \mathbf{X} = 2$  and

$$\begin{aligned} \mathbf{H} &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \frac{1}{2} \cdot \begin{bmatrix} 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}. \end{aligned}$$

Then  $\mathbf{H}\tilde{\mathbf{y}} = (\bar{y}, \bar{y})$ , where  $\bar{y} = (y_1 + y_2)/2$ : the fitted values of an intercept-only model are the [sample mean](#).

**Theorem 0.2** (Hat matrix is a projection matrix). *If  $\mathbf{X}$  is an  $n \times p$  design matrix with  $\text{rank}(\mathbf{X}) = p$ , then the hat matrix  $\mathbf{H}$  (Definition 0.2) is an orthogonal projection matrix (Definition 8 in [Special Matrices and Decompositions](#)).*

### **i** Proof

*Proof.* We verify symmetry and idempotency. Both use that  $\mathbf{X}^\top \mathbf{X}$  is symmetric,

$$\begin{aligned} (\mathbf{X}^\top \mathbf{X})^\top &= \mathbf{X}^\top (\mathbf{X}^\top)^\top \\ &= \mathbf{X}^\top \mathbf{X} \end{aligned}$$

([Theorem 16 in Matrices](#)), so its inverse is symmetric too ([Corollary 1 in Special Matrices and Decompositions](#)).

**Symmetry:**

$$\begin{aligned}
\mathbf{H}^\top &= (\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top)^\top && \text{(definition of } \mathbf{H} \text{)} \\
&= (\mathbf{X}^\top)^\top \cdot ((\mathbf{X}^\top \mathbf{X})^{-1})^\top \cdot \mathbf{X}^\top && \text{(transpose of a product, twice)} \\
&= \mathbf{X} \cdot ((\mathbf{X}^\top \mathbf{X})^{-1})^\top \cdot \mathbf{X}^\top && \text{(transposing twice changes nothing)} \\
&= \mathbf{X} \cdot (\mathbf{X}^\top \mathbf{X})^{-1} \cdot \mathbf{X}^\top && \text{(the inverse of a symmetric matrix is symmetric)} \\
&= \mathbf{H} && \text{(definition of } \mathbf{H} \text{)}
\end{aligned}$$

**Idempotency:**

$$\begin{aligned}
\mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \cdot \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of } \mathbf{H} \text{)} \\
&= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X}) (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(regroup; matrix multiplication is associative)} \\
&= \mathbf{X} \mathbf{I}_p (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(definition of the inverse)} \\
&= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top && \text{(identity matrix)} \\
&= \mathbf{H} && \text{(definition of } \mathbf{H} \text{)}
\end{aligned}$$

□

**Example 0.3** (The intercept-only hat matrix is a projection). For  $\mathbf{H} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$  from Example 0.2,  $\mathbf{H}^\top = \mathbf{H}$ , and

$$\begin{aligned}
\mathbf{H}^2 &= \begin{bmatrix} 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \\ 0.5 \cdot 0.5 + 0.5 \cdot 0.5 & 0.5 \cdot 0.5 + 0.5 \cdot 0.5 \end{bmatrix} \\
&= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \\
&= \mathbf{H},
\end{aligned}$$

as Theorem 0.2 says.

*Remark 0.2* (Why it is called the hat matrix). The hat matrix gives the fitted values in linear regression:

$$\begin{aligned}
\hat{\tilde{y}} &= \mathbf{X} \hat{\tilde{\beta}} \\
&= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \tilde{y} \\
&= \mathbf{H} \tilde{y}.
\end{aligned}$$

Multiplying by  $\mathbf{H}$  “puts a hat” on  $\tilde{y}$ , which is where the name comes from. In Example 0.2,  $\mathbf{H}$  puts a hat on  $\tilde{y}$  by replacing each  $y_i$  with the sample mean  $\bar{y}$ .

## Orthogonal projection onto a subspace

### Source

This section is adapted from the second half of Zhou (2024c), used under the MIT License (see the license text in [Section 2 in Subspaces and Rank](#)). It connects the orthogonal projection matrices of [Definition 8 in Special Matrices and Decompositions](#), defined by an algebraic property, to the geometric decomposition of [Theorem 31 in Subspaces and Rank](#).

**Definition 0.3** (Orthogonal projection onto a subspace). Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^p$  and  $\tilde{y} \in \mathbb{R}^p$ . By [Theorem 31 in Subspaces and Rank](#),  $\tilde{y} = \tilde{u} + \tilde{v}$  for exactly one  $\tilde{u} \in \mathcal{S}$  and  $\tilde{v} \in \mathcal{S}^\perp$ . The vector  $\tilde{u}$  is the **orthogonal projection** of  $\tilde{y}$  onto  $\mathcal{S}$ . It is the projection of  $\tilde{y}$  onto  $\mathcal{S}$  along  $\mathcal{S}^\perp$  ([Definition 18 in Subspaces and Rank](#)).

**Example 0.4** (Projecting onto a line in  $\mathbb{R}^3$ ). In [Example 56 in Subspaces and Rank](#),  $\tilde{y} = (3, 1, 2)$  splits as  $(2, 2, 0) + (1, -1, 2)$ , with  $(2, 2, 0) \in \mathcal{S} = \text{span}\{(1, 1, 0)\}$  and  $(1, -1, 2) \in \mathcal{S}^\perp$ . So the orthogonal projection of  $(3, 1, 2)$  onto  $\mathcal{S}$  is  $(2, 2, 0)$ . A vector already in  $\mathcal{S}$ , such as  $(5, 5, 0)$ , is its own projection, since  $(5, 5, 0) = (5, 5, 0) + \tilde{0}$  and  $\tilde{0} \in \mathcal{S}^\perp$  ([Theorem 29 in Subspaces and Rank](#), [Theorem 1 in Subspaces and Rank](#)); a vector of  $\mathcal{S}^\perp$ , such as  $(1, -1, 2)$ , projects to  $\tilde{0}$ , since  $(1, -1, 2) = \tilde{0} + (1, -1, 2)$  and  $\tilde{0} \in \mathcal{S}$  ([Theorem 1 in Subspaces and Rank](#)).

**Example 0.5** (A split that is not the orthogonal one).  $(3, 1, 2) = (3, 3, 0) + (0, -2, 2)$  also writes  $(3, 1, 2)$  as a vector of  $\mathcal{S} = \text{span}\{(1, 1, 0)\}$  plus a remainder, but  $(1, 1, 0) \cdot (0, -2, 2) = -2 \neq 0$ , so the remainder is not in  $\mathcal{S}^\perp$ , and  $(3, 3, 0)$  is not the orthogonal projection of  $(3, 1, 2)$  onto  $\mathcal{S}$ .

**Theorem 0.3** (The orthogonal projection is the closest point). *Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^p$ , let  $\tilde{y} \in \mathbb{R}^p$ , and let  $\tilde{u}$  be the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{S}$  ([Definition 0.3](#)). Then for every  $\tilde{w} \in \mathcal{S}$ ,*

$$\|\tilde{y} - \tilde{u}\| \leq \|\tilde{y} - \tilde{w}\|,$$

*with equality only when  $\tilde{w} = \tilde{u}$ .*

**i** Proof

*Proof.* By Definition 0.3,  $\tilde{y} - \tilde{u} \in \mathcal{S}^\perp$ . Both  $\tilde{u}$  and  $\tilde{w}$  are in  $\mathcal{S}$ , so  $\tilde{u} - \tilde{w} = \tilde{u} + (-1)\tilde{w} \in \mathcal{S}$  (Definition 4 in Subspaces and Rank), and therefore  $(\tilde{y} - \tilde{u}) \perp (\tilde{u} - \tilde{w})$  (Definition 19 in Subspaces and Rank). Then

$$\begin{aligned}\|\tilde{y} - \tilde{w}\|^2 &= \|(\tilde{y} - \tilde{u}) + (\tilde{u} - \tilde{w})\|^2 \quad (\text{add and subtract } \tilde{u}) \\ &= \|\tilde{y} - \tilde{u}\|^2 + \|\tilde{u} - \tilde{w}\|^2 \quad (\text{Pythagorean theorem, Theorem 2 in Inner Products and Orthogonality}) \\ &\geq \|\tilde{y} - \tilde{u}\|^2, \quad (\|\tilde{u} - \tilde{w}\|^2 \geq 0, \text{Theorem 1 in Inner Products and Orthogonality})\end{aligned}$$

with equality exactly when  $\|\tilde{u} - \tilde{w}\| = 0$ , that is, when  $\tilde{w} = \tilde{u}$  (Theorem 1 in Inner Products and Orthogonality, part 1). Taking nonnegative square roots preserves the inequality.  $\square$

**Example 0.6** (No point of the line is closer). In Example 0.4,  $\tilde{y} = (3, 1, 2)$  has projection  $\tilde{u} = (2, 2, 0)$  onto  $\mathcal{S} = \text{span}\{(1, 1, 0)\}$ , at distance  $\|(1, -1, 2)\| = \sqrt{6}$ . Another point of  $\mathcal{S}$ , such as  $\tilde{w} = (3, 3, 0)$ , is farther:

$$\begin{aligned}\|\tilde{y} - \tilde{w}\| &= \|(0, -2, 2)\| \\ &= \sqrt{8} \\ &> \sqrt{6},\end{aligned}$$

and the difference of squares is

$$\begin{aligned}\|\tilde{u} - \tilde{w}\|^2 &= \|(-1, -1, 0)\|^2 \\ &= 2 \\ &= 8 - 6,\end{aligned}$$

by the Pythagorean theorem (Theorem 2 in Inner Products and Orthogonality), since  $\tilde{y} - \tilde{u} \in \mathcal{S}^\perp$  and  $\tilde{u} - \tilde{w} \in \mathcal{S}$ .

**Theorem 0.4** (Projecting with an orthonormal basis). *Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^p$  with  $\dim(\mathcal{S}) = r \geq 1$ , and let  $\mathbf{Q}$  be the  $p \times r$  matrix whose columns  $\tilde{q}_1, \dots, \tilde{q}_r$  are an orthonormal basis of  $\mathcal{S}$  (Definition 9 in Inner Products and Orthogonality, Corollary 1 in Inner Products and Orthogonality). Then for every  $\tilde{y} \in \mathbb{R}^p$ ,  $\mathbf{Q}\mathbf{Q}^\top \tilde{y}$  is the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{S}$ , and  $\mathbf{Q}\mathbf{Q}^\top$  is an orthogonal projection matrix (Definition 8 in Special Matrices and Decompositions).*

**i** Proof

*Proof.*  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$ . Write  $q_{ki}$  for entry  $k$  of  $\tilde{q}_i$ , which is entry  $(k, i)$  of  $\mathbf{Q}$ . Then

$$\begin{aligned} (\mathbf{Q}^\top \mathbf{Q})_{ij} &= \sum_{k=1}^p (\mathbf{Q}^\top)_{ik} q_{kj} \quad (\text{Definition 7 in Matrices}) \\ &= \sum_{k=1}^p q_{ki} q_{kj} \quad (\text{Definition 3 in Matrices}) \\ &= \tilde{q}_i \cdot \tilde{q}_j, \quad (\text{Definition 7 in Vectors}) \end{aligned}$$

which is 1 if  $i = j$  and 0 otherwise (Definition 17 in Vectors), so  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$  (Definition 10 in Matrices).

**The split.** Let  $\tilde{u} = \mathbf{Q}\mathbf{Q}^\top \tilde{y}$  and  $\tilde{v} = \tilde{y} - \tilde{u}$ , so  $\tilde{y} = \tilde{u} + \tilde{v}$ .  $\tilde{u} = \mathbf{Q}(\mathbf{Q}^\top \tilde{y})$  (Theorem 5 in Matrices) is in  $\mathcal{C}(\mathbf{Q})$  (Definition 9 in Subspaces and Rank), which is  $\text{span}\{\tilde{q}_1, \dots, \tilde{q}_r\} = \mathcal{S}$  (Theorem 12 in Subspaces and Rank). For  $\tilde{v}$ ,

$$\begin{aligned} \mathbf{Q}^\top \tilde{v} &= \mathbf{Q}^\top (\tilde{y} - \mathbf{Q}\mathbf{Q}^\top \tilde{y}) \quad (\text{substitute}) \\ &= \mathbf{Q}^\top \tilde{y} - \mathbf{Q}^\top (\mathbf{Q}\mathbf{Q}^\top \tilde{y}) \quad (\text{Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \mathbf{Q}^\top \tilde{y} - (\mathbf{Q}^\top \mathbf{Q}) \mathbf{Q}^\top \tilde{y} \quad (\text{Theorem 5 in Matrices}) \\ &= \mathbf{Q}^\top \tilde{y} - \mathbf{I}_r \mathbf{Q}^\top \tilde{y} \quad (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r, \text{ from the first part of the proof}) \\ &= \mathbf{Q}^\top \tilde{y} - \mathbf{Q}^\top \tilde{y} \quad (\text{Theorem 7 in Matrices}) \\ &= \tilde{0}_r, \quad (\text{arithmetic}) \end{aligned}$$

so  $\tilde{v} \in \mathcal{N}(\mathbf{Q}^\top)$ , and

$$\begin{aligned} \mathcal{N}(\mathbf{Q}^\top) &= \mathcal{C}(\mathbf{Q})^\perp \\ &= \mathcal{S}^\perp \end{aligned}$$

(Theorem 30 in Subspaces and Rank). By the uniqueness in Theorem 31 in Subspaces and Rank,  $\tilde{u}$  is the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{S}$ .

**$\mathbf{Q}\mathbf{Q}^\top$  is symmetric and idempotent.**

$$\begin{aligned} (\mathbf{Q}\mathbf{Q}^\top)^\top &= (\mathbf{Q}^\top)^\top \mathbf{Q}^\top \quad (\text{Theorem 16 in Matrices}) \\ &= \mathbf{Q}\mathbf{Q}^\top, \quad (\text{transposing twice, Definition 3 in Matrices}) \end{aligned}$$

and

$$\begin{aligned} (\mathbf{Q}\mathbf{Q}^\top)(\mathbf{Q}\mathbf{Q}^\top) &= \mathbf{Q}(\mathbf{Q}^\top \mathbf{Q}) \mathbf{Q}^\top \quad (\text{Theorem 5 in Matrices}) \\ &= \mathbf{Q}\mathbf{I}_r \mathbf{Q}^\top \quad (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r, \text{ from the first part of the proof}) \\ &= \mathbf{Q}\mathbf{Q}^\top. \quad (\text{Theorem 7 in Matrices}) \end{aligned}$$

□

**Example 0.7** (The projection matrix onto a line). For  $\mathcal{S} = \text{span}\{(1, 1, 0)\}$ , the single vector  $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$  is an orthonormal basis: it spans  $\mathcal{S}$ , it is nonzero and so linearly independent, and

$$\begin{aligned}\|\tilde{q}_1\| &= \sqrt{\frac{1}{2} + \frac{1}{2} + 0} \\ &= 1.\end{aligned}$$

With  $\mathbf{Q} = [\tilde{q}_1]$ ,

$$\begin{aligned}\mathbf{Q}\mathbf{Q}^\top &= \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{bmatrix}.\end{aligned}$$

Applied to  $\tilde{y} = (3, 1, 2)$ , it gives  $(\frac{3+1}{2}, \frac{3+1}{2}, 0) = (2, 2, 0)$ , the projection found in Example 0.4.

**Theorem 0.5** (A projection matrix projects onto its column space, and is the only one that does).

1. If  $\mathbf{P}$  is a  $p \times p$  orthogonal projection matrix ([Definition 8 in Special Matrices and Decompositions](#)), then for every  $\tilde{y} \in \mathbb{R}^p$ ,  $\mathbf{P}\tilde{y}$  is the orthogonal projection of  $\tilde{y}$  onto  $\mathcal{C}(\mathbf{P})$  ([Definition 0.3](#)).
2. If  $\mathbf{P}_1$  and  $\mathbf{P}_2$  are  $p \times p$  matrices such that, for every  $\tilde{y}$ , both  $\mathbf{P}_1\tilde{y}$  and  $\mathbf{P}_2\tilde{y}$  are the orthogonal projection of  $\tilde{y}$  onto the same subspace  $\mathcal{S}$ , then  $\mathbf{P}_1 = \mathbf{P}_2$ .

### Proof

*Proof. Part 1.* Write  $\tilde{y} = \mathbf{P}\tilde{y} + (\tilde{y} - \mathbf{P}\tilde{y})$ . The first term is in  $\mathcal{C}(\mathbf{P})$  ([Definition 9 in Subspaces and Rank](#)). For the second,

$$\begin{aligned}\mathbf{P}(\tilde{y} - \mathbf{P}\tilde{y}) &= \mathbf{P}\tilde{y} - \mathbf{P}(\mathbf{P}\tilde{y}) \quad (\text{Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \mathbf{P}\tilde{y} - (\mathbf{P}\mathbf{P})\tilde{y} \quad (\text{Theorem 5 in Matrices}) \\ &= \mathbf{P}\tilde{y} - \mathbf{P}^2\tilde{y} \quad (\text{Definition 2 in Special Matrices and Decompositions}) \\ &= \mathbf{P}\tilde{y} - \mathbf{P}\tilde{y} \quad (\mathbf{P} \text{ is idempotent}) \\ &= \tilde{0}_p, \quad (\text{arithmetic})\end{aligned}$$

so  $\tilde{y} - \mathbf{P}\tilde{y} \in \mathcal{N}(\mathbf{P})$ . Since  $\mathbf{P} = \mathbf{P}^\top$ ,

$$\begin{aligned}\mathcal{N}(\mathbf{P}) &= \mathcal{N}(\mathbf{P}^\top) \\ &= \mathcal{C}(\mathbf{P})^\perp\end{aligned}$$

(Theorem 30 in Subspaces and Rank). By the uniqueness in Theorem 31 in Subspaces and Rank,  $\mathbf{P}\tilde{\mathbf{y}}$  is the orthogonal projection of  $\tilde{\mathbf{y}}$  onto  $\mathcal{C}(\mathbf{P})$ .

**Part 2.** The orthogonal projection of a vector onto  $\mathcal{S}$  is unique (Definition 0.3), so  $\mathbf{P}_1\tilde{\mathbf{e}}_j = \mathbf{P}_2\tilde{\mathbf{e}}_j$  for each indicator vector  $\tilde{\mathbf{e}}_j$  (Definition 12 in Vectors).  $\mathbf{P}_1\tilde{\mathbf{e}}_j$  is column  $j$  of  $\mathbf{P}_1$ : by Theorem 3 in Subspaces and Rank it is the combination of the columns of  $\mathbf{P}_1$  with coefficients the entries of  $\tilde{\mathbf{e}}_j$ , which are 1 in place  $j$  and 0 elsewhere (Definition 12 in Vectors). Likewise for  $\mathbf{P}_2$ , so the two matrices have the same columns.  $\square$

**Example 0.8** (Two routes to the same projection matrix). The matrix  $\mathbf{P} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  of Example 10 in Special Matrices and Decompositions is an orthogonal projection matrix, and its column space is  $\text{span}\{(1, 0)\}$ , the horizontal axis. By part 1,  $\mathbf{P}(v_1, v_2) = (v_1, 0)$  is the orthogonal projection onto that axis: the remainder  $(0, v_2)$  is orthogonal to  $(1, 0)$ . The vector  $\tilde{\mathbf{q}}_1 = (1, 0)$  has length 1 and spans the axis, so it is an orthonormal basis of it, and with  $\mathbf{B} = [\tilde{\mathbf{q}}_1]$ ,

$$\begin{aligned}\mathbf{B}\mathbf{B}^\top &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \\ &= \mathbf{P},\end{aligned}$$

as part 2 and Theorem 0.4 require.

The oblique projection  $\mathbf{Q} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$  of Example 10 in Special Matrices and Decompositions also maps every vector into the horizontal axis. It sends  $(0, 1)$  to  $(1, 0)$ , but the remainder  $(0, 1) - (1, 0) = (-1, 1)$  is not orthogonal to the axis, so  $(1, 0)$  is not the orthogonal projection of  $(0, 1)$  onto the axis, which is  $\mathbf{P}(0, 1) = (0, 0)$ .

**Theorem 0.6** (The hat matrix projects onto the column space of the design matrix). *If  $\mathbf{X}$  is an  $n \times p$  design matrix with  $\text{rank}(\mathbf{X}) = p$ , then for every  $\tilde{\mathbf{y}} \in \mathbb{R}^n$ ,  $\mathbf{H}\tilde{\mathbf{y}}$  (Definition 0.2) is the orthogonal projection of  $\tilde{\mathbf{y}}$  onto  $\mathcal{C}(\mathbf{X})$ . So  $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$  for any  $n \times p$  matrix  $\mathbf{Q}$  whose columns are an orthonormal basis of  $\mathcal{C}(\mathbf{X})$ .*

### **i** Proof

*Proof.*  $\mathbf{H}$  is an orthogonal projection matrix (Theorem 0.2), so  $\mathbf{H}\tilde{\mathbf{y}}$  is the orthogonal projection of  $\tilde{\mathbf{y}}$  onto  $\mathcal{C}(\mathbf{H})$  (Theorem 0.5, part 1). It remains to show  $\mathcal{C}(\mathbf{H}) = \mathcal{C}(\mathbf{X})$ .  $\mathcal{C}(\mathbf{H}) \subseteq \mathcal{C}(\mathbf{X})$ .  $\mathbf{H}\tilde{\mathbf{z}} = \mathbf{X}((\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\tilde{\mathbf{z}})$  (Theorem 5 in Matrices), which is in  $\mathcal{C}(\mathbf{X})$

(Definition 9 in Subspaces and Rank).  
 $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{H})$ . First,

$$\begin{aligned}
\mathbf{H}\mathbf{X} &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} && \text{(Definition 0.2)} \\
&= \mathbf{X} ((\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{X}^\top \mathbf{X})) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{X} \mathbf{I}_p && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{X}. && \text{(Theorem 7 in Matrices)}
\end{aligned}$$

So for every  $\tilde{b} \in \mathbb{R}^p$ ,  $\mathbf{X}\tilde{b} = (\mathbf{H}\mathbf{X})\tilde{b}$ , since  $\mathbf{H}\mathbf{X} = \mathbf{X}$ , and  $(\mathbf{H}\mathbf{X})\tilde{b} = \mathbf{H}(\mathbf{X}\tilde{b})$  (Theorem 5 in Matrices), which is in  $\mathcal{C}(\mathbf{H})$ .

$\mathcal{C}(\mathbf{X})$  has dimension  $p \geq 1$  (Theorem 16 in Subspaces and Rank), so an orthonormal basis of it has  $p$  vectors (Definition 8 in Subspaces and Rank), and  $\mathbf{Q}\mathbf{Q}^\top$  also gives the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$  (Theorem 0.4, with  $n$  in place of  $p$  and  $r = p$ ). By Theorem 0.5, part 2,  $\mathbf{H} = \mathbf{Q}\mathbf{Q}^\top$ .  $\square$

**Example 0.9** (The intercept-only hat matrix, from an orthonormal basis). For the intercept-only design  $\mathbf{X} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  of Example 0.2,  $\mathcal{C}(\mathbf{X}) = \text{span}\{(1, 1)\}$ , with orthonormal basis  $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1, 1)$ , since

$$\begin{aligned}
\|\tilde{q}_1\| &= \sqrt{\frac{1}{2} + \frac{1}{2}} \\
&= 1.
\end{aligned}$$

With  $\mathbf{Q} = [\tilde{q}_1]$ ,

$$\begin{aligned}
\mathbf{Q}\mathbf{Q}^\top &= \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} \\
&= \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix},
\end{aligned}$$

the hat matrix found in Example 0.2. The fitted values  $(\bar{y}, \bar{y})$  are the closest point to  $\tilde{y}$  on the line of constant vectors (Theorem 0.3), and the residuals  $(y_1 - \bar{y}, y_2 - \bar{y})$  are orthogonal to  $(1, 1)$ :

$$\begin{aligned}
(1, 1) \cdot (y_1 - \bar{y}, y_2 - \bar{y}) &= y_1 + y_2 - 2\bar{y} \\
&= 0,
\end{aligned}$$

since  $2\bar{y} = y_1 + y_2$ .

## Generalized inverses and linear systems

### i Source

This section is adapted from Zhou (2024b), used under the MIT License (see the license text in [Section 2 in Subspaces and Rank](#)). The source defers the existence of the Moore-Penrose inverse to the singular value decomposition; the proof here builds it from a rank factorization ([Theorem 23 in Subspaces and Rank](#)) instead, and the section adds the projection  $\mathbf{XGX}^\top$  for any generalized inverse  $\mathbf{G}$  of  $\mathbf{X}^\top \mathbf{X}$ , which the source does not cover.

**Theorem 0.7** (A scalar factor passes through a matrix product). *For an  $m \times k$  matrix  $\mathbf{A}$ , a  $k \times n$  matrix  $\mathbf{B}$  and a number  $c$ ,*

$$\begin{aligned}\mathbf{A}(c\mathbf{B}) &= c(\mathbf{AB}) \\ &= (c\mathbf{A})\mathbf{B}.\end{aligned}$$

*In particular,  $\mathbf{A}(\mathbf{B} - \mathbf{C}) = \mathbf{AB} - \mathbf{AC}$  and  $(\mathbf{B} - \mathbf{C})\mathbf{D} = \mathbf{BD} - \mathbf{CD}$  for matrices of compatible sizes, where  $\mathbf{B} - \mathbf{C} = \mathbf{B} + (-1)\mathbf{C}$  ([Theorem 4 in Matrices](#), [Definition 6 in Matrices](#)).*

### i Proof

*Proof.*  $\mathbf{A}(c\mathbf{B}) = c(\mathbf{AB})$ . For each entry  $(i, j)$ ,

$$\begin{aligned}(\mathbf{A}(c\mathbf{B}))_{ij} &= \sum_{l=1}^k a_{il}(c\mathbf{B})_{lj} \quad (\text{Definition 7 in Matrices}) \\ &= \sum_{l=1}^k a_{il}(cb_{lj}) \quad (\text{Definition 6 in Matrices}) \\ &= \sum_{l=1}^k c(a_{il}b_{lj}) \quad (\text{reorder the product of numbers}) \\ &= c \sum_{l=1}^k a_{il}b_{lj} \quad (\text{factor } c \text{ out of the sum}) \\ &= c(\mathbf{AB})_{ij} \quad (\text{Definition 7 in Matrices}) \\ &= (c(\mathbf{AB}))_{ij}. \quad (\text{Definition 6 in Matrices})\end{aligned}$$

$(c\mathbf{A})\mathbf{B} = c(\mathbf{AB})$ . For each entry  $(i, j)$ ,

$$\begin{aligned}
((c \mathbf{A}) \mathbf{B})_{ij} &= \sum_{l=1}^k (c a_{il}) b_{lj} \quad (\text{Definition 7 in Matrices, Definition 6 in Matrices}) \\
&= c \sum_{l=1}^k a_{il} b_{lj} \quad (\text{factor } c \text{ out of the sum}) \\
&= (c (\mathbf{AB}))_{ij}. \quad (\text{Definition 7 in Matrices, Definition 6 in Matrices})
\end{aligned}$$

**Differences.**

$$\begin{aligned}
\mathbf{A} (\mathbf{B} - \mathbf{C}) &= \mathbf{A} (\mathbf{B} + (-1) \mathbf{C}) \quad (\text{the difference as a sum}) \\
&= \mathbf{AB} + \mathbf{A} ((-1) \mathbf{C}) \quad (\text{Theorem 6 in Matrices}) \\
&= \mathbf{AB} + (-1) (\mathbf{AC}) \quad (\text{first part}) \\
&= \mathbf{AB} - \mathbf{AC}, \quad (\text{the difference as a sum})
\end{aligned}$$

and  $(\mathbf{B} - \mathbf{C}) \mathbf{D} = \mathbf{BD} - \mathbf{CD}$  the same way, using the second law in [Theorem 6 in Matrices](#) and the second part. A column vector is a matrix with one column, so all of this applies with  $\mathbf{B}$ ,  $\mathbf{C}$  or  $\mathbf{D}$  a vector.  $\square$

**Example 0.10** (Pulling a scalar out of a product). With  $\mathbf{A} = \begin{bmatrix} 1 & 2 \end{bmatrix}$  and  $\mathbf{B} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ :

$$\begin{aligned}
\mathbf{A} (2 \mathbf{B}) &= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 8 \end{bmatrix} \\
&= 22,
\end{aligned}$$

and

$$\begin{aligned}
2 (\mathbf{AB}) &= 2 (3 + 8) \\
&= 22.
\end{aligned}$$

**Definition 0.4** (Generalized inverse). A **generalized inverse** of an  $m \times n$  matrix  $\mathbf{A}$  is an  $n \times m$  matrix  $\mathbf{G}$  such that

$$\underbrace{\mathbf{A}}_{m \times n} \underbrace{\mathbf{G}}_{n \times m} \underbrace{\mathbf{A}}_{m \times n} = \mathbf{A}.$$

**Example 0.11** (A matrix with many generalized inverses). Let  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  and  $\mathbf{G} =$

$$\begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix}. \text{ Then}$$

$$\begin{aligned}\mathbf{A}\mathbf{G}\mathbf{A} &= \begin{bmatrix} g_{11} & g_{12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad (\text{Definition 7 in Matrices, for } \mathbf{A}\mathbf{G}) \\ &= \begin{bmatrix} g_{11} & 0 \\ 0 & 0 \end{bmatrix}, \quad (\text{Definition 7 in Matrices})\end{aligned}$$

which equals  $\mathbf{A}$  exactly when  $g_{11} = 1$ . So every matrix  $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$  is a generalized inverse of  $\mathbf{A}$ : a generalized inverse need not be unique. The zero matrix is not one, since its  $(1, 1)$  entry is 0.

**Theorem 0.8** (An invertible matrix has only one generalized inverse). *If  $\mathbf{A}$  is an invertible  $p \times p$  matrix (Definition 6 in Special Matrices and Decompositions), its only generalized inverse is  $\mathbf{A}^{-1}$ .*

**i** Proof

*Proof.*

$$\begin{aligned}\mathbf{A}\mathbf{A}^{-1}\mathbf{A} &= \mathbf{I}_p\mathbf{A} \\ &= \mathbf{A}\end{aligned}$$

(Definition 5 in Special Matrices and Decompositions, Theorem 7 in Matrices), so  $\mathbf{A}^{-1}$  is a generalized inverse. If  $\mathbf{G}$  is any generalized inverse, then

$$\begin{aligned}\mathbf{G} &= \mathbf{I}_p\mathbf{G}\mathbf{I}_p && (\text{Theorem 7 in Matrices}) \\ &= (\mathbf{A}^{-1}\mathbf{A})\mathbf{G}(\mathbf{A}\mathbf{A}^{-1}) && (\text{Definition 5 in Special Matrices and Decompositions}) \\ &= \mathbf{A}^{-1}(\mathbf{A}\mathbf{G}\mathbf{A})\mathbf{A}^{-1} && (\text{Theorem 5 in Matrices}) \\ &= \mathbf{A}^{-1}\mathbf{A}\mathbf{A}^{-1} && (\text{Definition 0.4}) \\ &= \mathbf{I}_p\mathbf{A}^{-1} && (\text{Definition 5 in Special Matrices and Decompositions}) \\ &= \mathbf{A}^{-1}. && (\text{Theorem 7 in Matrices})\end{aligned}$$

□

**Example 0.12** (The generalized inverse of an invertible matrix). For  $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$  of Example 5 in Special Matrices and Decompositions, the only generalized inverse is  $\mathbf{A}^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ 0 & 1 \end{bmatrix}$ . The matrix of Example 0.11, by contrast, has infinitely many, so by Theorem 0.8 it is singular.

**Theorem 0.9** (A generalized inverse solves every solvable system). *Let  $\mathbf{A}$  be  $m \times n$ , let  $\mathbf{G}$  be a generalized inverse of  $\mathbf{A}$  (Definition 0.4), and let  $\tilde{\mathbf{b}} \in \mathbb{R}^m$ . If  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  has a solution, then  $\mathbf{G}\tilde{\mathbf{b}}$  is a solution.*

### Proof

*Proof.* Let  $\tilde{\mathbf{x}}_0$  be a solution, so  $\mathbf{A}\tilde{\mathbf{x}}_0 = \tilde{\mathbf{b}}$ . Then

$$\begin{aligned} \mathbf{A}(\mathbf{G}\tilde{\mathbf{b}}) &= \mathbf{A}\mathbf{G}(\mathbf{A}\tilde{\mathbf{x}}_0) && \text{(substitute } \tilde{\mathbf{b}} = \mathbf{A}\tilde{\mathbf{x}}_0\text{)} \\ &= (\mathbf{A}\mathbf{G}\mathbf{A})\tilde{\mathbf{x}}_0 && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{A}\tilde{\mathbf{x}}_0 && \text{(Definition 0.4)} \\ &= \tilde{\mathbf{b}}. && (\tilde{\mathbf{x}}_0 \text{ is a solution}) \end{aligned}$$

□

**Example 0.13** (Solving with a generalized inverse). With  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  and its generalized inverse  $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$  (Example 0.11), the system  $\mathbf{A}\tilde{\mathbf{x}} = (2, 0)$  has the solution  $(2, 0)$ , and  $\mathbf{G}(2, 0) = (2, 14)$  is a solution too:  $\mathbf{A}(2, 14) = (2, 0)$ . The system  $\mathbf{A}\tilde{\mathbf{x}} = (2, 1)$  has no solution, since the second entry of  $\mathbf{A}\tilde{\mathbf{x}}$  is always 0; there  $\mathbf{G}(2, 1) = (7, 14)$  gives  $\mathbf{A}(7, 14) = (7, 0) \neq (2, 1)$ .

**Definition 0.5** (Moore-Penrose inverse). A **Moore-Penrose inverse** of an  $m \times n$  matrix  $\mathbf{A}$  is an  $n \times m$  matrix  $\mathbf{G}$  satisfying all four conditions

1.  $\mathbf{A}\mathbf{G}\mathbf{A} = \mathbf{A}$ ;
2.  $\mathbf{G}\mathbf{A}\mathbf{G} = \mathbf{G}$ ;
3.  $(\mathbf{A}\mathbf{G})^\top = \mathbf{A}\mathbf{G}$ ;
4.  $(\mathbf{G}\mathbf{A})^\top = \mathbf{G}\mathbf{A}$ .

**Example 0.14** (Checking the four conditions). For  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ , the matrix  $\mathbf{G} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  satisfies all four:

$$\begin{aligned} \mathbf{A}\mathbf{G} &= \mathbf{G}\mathbf{A} \\ &= \mathbf{A} \end{aligned}$$

and  $\mathbf{A}^2 = \mathbf{A}$ , so conditions 1 and 2 read  $\mathbf{A} = \mathbf{A}$ , and conditions 3 and 4 hold because  $\mathbf{A}$  is symmetric.

The generalized inverse  $\mathbf{G}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$  of  $\mathbf{A}$  (Example 0.11) is not a Moore-Penrose inverse:  $\mathbf{A}\mathbf{G}' = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ , which is not symmetric, so condition 3 fails.

**Theorem 0.10** (A Moore-Penrose inverse is a generalized inverse). *If  $\mathbf{G}$  is a Moore-Penrose inverse of an  $m \times n$  matrix  $\mathbf{A}$  (Definition 0.5), then  $\mathbf{G}$  is a generalized inverse of  $\mathbf{A}$  (Definition 0.4).*

### **i** Proof

*Proof.* A generalized inverse of  $\mathbf{A}$  is an  $n \times m$  matrix  $\mathbf{G}$  with  $\mathbf{A}\mathbf{G}\mathbf{A} = \mathbf{A}$ . A Moore-Penrose inverse is an  $n \times m$  matrix, and condition 1 of Definition 0.5 is  $\mathbf{A}\mathbf{G}\mathbf{A} = \mathbf{A}$ .  $\square$

**Example 0.15** (The Moore-Penrose inverse of a rank-one matrix). The  $3 \times 2$  matrix  $\mathbf{A}$  below has rank 1, because its second column is twice its first, so it has no ordinary inverse. The function `MASS::ginv()` computes its Moore-Penrose inverse  $\mathbf{G}$ . The code checks the four conditions of Definition 0.5, of which the first is the equation that makes  $\mathbf{G}$  a generalized inverse.

```
A <- cbind(c(1, 2, 3), c(2, 4, 6))
G <- MASS::ginv(A)
conditions <- c(
  "1: AGA = A" = isTRUE(all.equal(A %*% G %*% A, A)),
  "2: GAG = G" = isTRUE(all.equal(G %*% A %*% G, G)),
  "3: AG symmetric" = isTRUE(all.equal(A %*% G, t(A %*% G))),
  "4: GA symmetric" = isTRUE(all.equal(G %*% A, t(G %*% A)))
)
conditions
#>      1: AGA = A      2: GAG = G 3: AG symmetric 4: GA symmetric
#>           TRUE           TRUE           TRUE           TRUE
round(G, 4)
#>      [,1]  [,2]  [,3]
#> [1,] 0.0143 0.0286 0.0429
#> [2,] 0.0286 0.0571 0.0857
```

All 4 of 4 conditions hold, so  $\mathbf{G}$  is a Moore-Penrose inverse of  $\mathbf{A}$ . By Theorem 0.10 it is also a generalized inverse, which is condition 1:  $\mathbf{A}\mathbf{G}\mathbf{A} = \mathbf{A}$  up to rounding error.

**Theorem 0.11** (A matrix has at most one Moore-Penrose inverse). *If  $\mathbf{G}_1$  and  $\mathbf{G}_2$  are both Moore-Penrose inverses of an  $m \times n$  matrix  $\mathbf{A}$  (Definition 0.5), then  $\mathbf{G}_1 = \mathbf{G}_2$ .*

**i** Proof

*Proof.*  $\mathbf{A}\mathbf{G}_1 = \mathbf{A}\mathbf{G}_2$ .

$$\begin{aligned}
 \mathbf{A}\mathbf{G}_1 &= (\mathbf{A}\mathbf{G}_1)^\top && \text{(condition 3 for } \mathbf{G}_1) \\
 &= \mathbf{G}_1^\top \mathbf{A}^\top && \text{(Theorem 16 in Matrices)} \\
 &= \mathbf{G}_1^\top (\mathbf{A}\mathbf{G}_2\mathbf{A})^\top && \text{(condition 1 for } \mathbf{G}_2) \\
 &= \mathbf{G}_1^\top \mathbf{A}^\top \mathbf{G}_2^\top \mathbf{A}^\top && \text{(Theorem 16 in Matrices, twice)} \\
 &= (\mathbf{G}_1^\top \mathbf{A}^\top) (\mathbf{G}_2^\top \mathbf{A}^\top) && \text{(Theorem 5 in Matrices)} \\
 &= (\mathbf{A}\mathbf{G}_1)^\top (\mathbf{A}\mathbf{G}_2)^\top && \text{(Theorem 16 in Matrices, twice)} \\
 &= (\mathbf{A}\mathbf{G}_1)(\mathbf{A}\mathbf{G}_2) && \text{(condition 3 for } \mathbf{G}_1 \text{ and } \mathbf{G}_2) \\
 &= (\mathbf{A}\mathbf{G}_1\mathbf{A})\mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
 &= \mathbf{A}\mathbf{G}_2. && \text{(condition 1 for } \mathbf{G}_1)
 \end{aligned}$$

$\mathbf{G}_1\mathbf{A} = \mathbf{G}_2\mathbf{A}$ .

$$\begin{aligned}
 \mathbf{G}_1\mathbf{A} &= (\mathbf{G}_1\mathbf{A})^\top && \text{(condition 4 for } \mathbf{G}_1) \\
 &= \mathbf{A}^\top \mathbf{G}_1^\top && \text{(Theorem 16 in Matrices)} \\
 &= (\mathbf{A}\mathbf{G}_2\mathbf{A})^\top \mathbf{G}_1^\top && \text{(condition 1 for } \mathbf{G}_2) \\
 &= \mathbf{A}^\top \mathbf{G}_2^\top \mathbf{A}^\top \mathbf{G}_1^\top && \text{(Theorem 16 in Matrices, twice)} \\
 &= (\mathbf{A}^\top \mathbf{G}_2^\top) (\mathbf{A}^\top \mathbf{G}_1^\top) && \text{(Theorem 5 in Matrices)} \\
 &= (\mathbf{G}_2\mathbf{A})^\top (\mathbf{G}_1\mathbf{A})^\top && \text{(Theorem 16 in Matrices, twice)} \\
 &= (\mathbf{G}_2\mathbf{A})(\mathbf{G}_1\mathbf{A}) && \text{(condition 4 for } \mathbf{G}_2 \text{ and } \mathbf{G}_1) \\
 &= \mathbf{G}_2(\mathbf{A}\mathbf{G}_1\mathbf{A}) && \text{(Theorem 5 in Matrices)} \\
 &= \mathbf{G}_2\mathbf{A}. && \text{(condition 1 for } \mathbf{G}_1)
 \end{aligned}$$

**Conclusion.**

$$\begin{aligned}
\mathbf{G}_1 &= \mathbf{G}_1 \mathbf{A} \mathbf{G}_1 && \text{(condition 2 for } \mathbf{G}_1) \\
&= \mathbf{G}_1 (\mathbf{A} \mathbf{G}_1) && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{G}_1 (\mathbf{A} \mathbf{G}_2) && \text{(the first step)} \\
&= (\mathbf{G}_1 \mathbf{A}) \mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{G}_2 \mathbf{A}) \mathbf{G}_2 && \text{(the second step)} \\
&= \mathbf{G}_2 \mathbf{A} \mathbf{G}_2 && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{G}_2. && \text{(condition 2 for } \mathbf{G}_2)
\end{aligned}$$

□

**Example 0.16** (Only one of the many generalized inverses qualifies). The matrix  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  has infinitely many generalized inverses  $\begin{bmatrix} 1 & g_{12} \\ g_{21} & g_{22} \end{bmatrix}$  (Example 0.11), and  $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  is a Moore-Penrose inverse (Example 0.14). By Theorem 0.11 it is the only one, so every other choice of  $g_{12}, g_{21}, g_{22}$  breaks condition 2, 3 or 4. For instance,  $g_{21} = 1$  (the rest 0) gives  $\mathbf{GA} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ , which is not symmetric.

**Theorem 0.12** (Every matrix has a Moore-Penrose inverse). *Every  $m \times n$  matrix  $\mathbf{A}$  has a Moore-Penrose inverse (Definition 0.5), written  $\mathbf{A}^+$ ; it is unique by Theorem 0.11.*

- If  $\mathbf{A} = \mathbf{0}_{m \times n}$ , then  $\mathbf{A}^+ = \mathbf{0}_{n \times m}$ .
- If  $\text{rank}(\mathbf{A}) = r \geq 1$  and  $\mathbf{A} = \mathbf{CR}$  is a rank factorization (Definition 15 in Subspaces and Rank), then

$$\underbrace{\mathbf{A}^+}_{n \times m} = \underbrace{\mathbf{R}^\top}_{n \times r} \underbrace{(\mathbf{R}\mathbf{R}^\top)^{-1}}_{r \times r} \underbrace{(\mathbf{C}^\top \mathbf{C})^{-1}}_{r \times r} \underbrace{\mathbf{C}^\top}_{r \times m}.$$

### **i** Proof

*Proof.* **The two cases cover every matrix.** If  $\mathbf{A} \neq \mathbf{0}_{m \times n}$ , it has a nonzero column, which is linearly independent on its own ( $c\tilde{v} = \tilde{0}$  with  $\tilde{v} \neq \tilde{0}$  forces  $c = 0$ ), so  $\text{rank}(\mathbf{A}) \geq 1$  (Definition 2 in Subspaces and Rank).

**The zero matrix.** With  $\mathbf{A} = \mathbf{0}_{m \times n}$  and  $\mathbf{G} = \mathbf{0}_{n \times m}$ , every product in conditions 1 to 4 is a zero matrix, and a zero matrix equals its own transpose, so all four conditions hold.

**The two inverses exist.** Now suppose  $\text{rank}(\mathbf{A}) = r \geq 1$ . A rank factorization exists (Theorem 23 in Subspaces and Rank).  $\mathbf{C}$  is  $m \times r$  and  $\mathbf{R}$  is  $r \times n$ . Both have rank  $r$ :

$$\begin{aligned}
r &= \text{rank}(\mathbf{C}\mathbf{R}) \quad (\mathbf{A} = \mathbf{C}\mathbf{R}) \\
&\leq \text{rank}(\mathbf{C}) \quad (\text{Theorem 18 in Subspaces and Rank}) \\
&\leq r, \quad (\text{Corollary 1 in Subspaces and Rank, } \mathbf{C} \text{ has } r \text{ columns})
\end{aligned}$$

and, using  $(\mathbf{C}\mathbf{R})^\top = \mathbf{R}^\top \mathbf{C}^\top$  (Theorem 16 in Matrices),

$$\begin{aligned}
r &= \text{rank}(\mathbf{R}^\top \mathbf{C}^\top) \quad (\text{Theorem 19 in Subspaces and Rank}) \\
&\leq \text{rank}(\mathbf{R}^\top) \quad (\text{Theorem 18 in Subspaces and Rank}) \\
&\leq r. \quad (\text{Corollary 1 in Subspaces and Rank, } \mathbf{R}^\top \text{ has } r \text{ columns})
\end{aligned}$$

So  $\mathbf{C}$  and  $\mathbf{R}^\top$  have full column rank  $r$ , and  $\mathbf{C}^\top \mathbf{C}$  and  $\mathbf{R}\mathbf{R}^\top = (\mathbf{R}^\top)^\top \mathbf{R}^\top$  are invertible (Theorem 0.1). They are symmetric:

$$\begin{aligned}
(\mathbf{C}^\top \mathbf{C})^\top &= \mathbf{C}^\top (\mathbf{C}^\top)^\top \\
&= \mathbf{C}^\top \mathbf{C}
\end{aligned}$$

(Theorem 16 in Matrices, Definition 3 in Matrices), and likewise for  $\mathbf{R}\mathbf{R}^\top$ . Write  $\mathbf{S} \stackrel{\text{def}}{=} (\mathbf{R}\mathbf{R}^\top)^{-1}$  and  $\mathbf{T} \stackrel{\text{def}}{=} (\mathbf{C}^\top \mathbf{C})^{-1}$ , both symmetric (Corollary 1 in Special Matrices and Decompositions), and  $\mathbf{G} \stackrel{\text{def}}{=} \mathbf{R}^\top \mathbf{S} \mathbf{T} \mathbf{C}^\top$ .

**Two products.**

$$\begin{aligned}
\mathbf{A}\mathbf{G} &= \mathbf{C}\mathbf{R}\mathbf{R}^\top \mathbf{S} \mathbf{T} \mathbf{C}^\top \quad (\text{substitute } \mathbf{A} \text{ and } \mathbf{G}) \\
&= \mathbf{C}(\mathbf{R}\mathbf{R}^\top) \mathbf{S} \mathbf{T} \mathbf{C}^\top \quad (\text{Theorem 5 in Matrices}) \\
&= \mathbf{C}\mathbf{I}_r \mathbf{T} \mathbf{C}^\top \quad (\text{Definition 5 in Special Matrices and Decompositions}) \\
&= \mathbf{C} \mathbf{T} \mathbf{C}^\top, \quad (\text{Theorem 7 in Matrices})
\end{aligned}$$

$$\begin{aligned}
\mathbf{G}\mathbf{A} &= \mathbf{R}^\top \mathbf{S} \mathbf{T} \mathbf{C}^\top \mathbf{C} \mathbf{R} \quad (\text{substitute } \mathbf{G} \text{ and } \mathbf{A}) \\
&= \mathbf{R}^\top \mathbf{S} \mathbf{T} (\mathbf{C}^\top \mathbf{C}) \mathbf{R} \quad (\text{Theorem 5 in Matrices}) \\
&= \mathbf{R}^\top \mathbf{S} \mathbf{I}_r \mathbf{R} \quad (\text{Definition 5 in Special Matrices and Decompositions}) \\
&= \mathbf{R}^\top \mathbf{S} \mathbf{R}. \quad (\text{Theorem 7 in Matrices})
\end{aligned}$$

**Conditions 3 and 4.**

$$\begin{aligned}
(\mathbf{A}\mathbf{G})^\top &= (\mathbf{C} \mathbf{T} \mathbf{C}^\top)^\top \quad (\text{first product}) \\
&= (\mathbf{C} (\mathbf{T} \mathbf{C}^\top))^\top \quad (\text{Theorem 5 in Matrices}) \\
&= (\mathbf{T} \mathbf{C}^\top)^\top \mathbf{C}^\top \quad (\text{Theorem 16 in Matrices}) \\
&= (\mathbf{C}^\top)^\top \mathbf{T}^\top \mathbf{C}^\top \quad (\text{Theorem 16 in Matrices}) \\
&= \mathbf{C} \mathbf{T}^\top \mathbf{C}^\top \quad (\text{Definition 3 in Matrices}) \\
&= \mathbf{C} \mathbf{T} \mathbf{C}^\top, \quad (\mathbf{T} \text{ is symmetric})
\end{aligned}$$

$$\begin{aligned}
(\mathbf{GA})^\top &= (\mathbf{R}^\top \mathbf{SR})^\top && \text{(second product)} \\
&= (\mathbf{R}^\top (\mathbf{SR}))^\top && \text{(Theorem 5 in Matrices)} \\
&= (\mathbf{SR})^\top (\mathbf{R}^\top)^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{S}^\top (\mathbf{R}^\top)^\top && \text{(Theorem 16 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{S}^\top \mathbf{R} && \text{(Definition 3 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{SR}. && (\mathbf{S} \text{ is symmetric})
\end{aligned}$$

**Condition 1.**

$$\begin{aligned}
\mathbf{AGA} &= \mathbf{CTC}^\top \mathbf{CR} && \text{(first product)} \\
&= \mathbf{CT}(\mathbf{C}^\top \mathbf{C}) \mathbf{R} && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{CI}_r \mathbf{R} && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{A}. && \text{(Theorem 7 in Matrices)}
\end{aligned}$$

**Condition 2.**

$$\begin{aligned}
\mathbf{GAG} &= \mathbf{R}^\top \mathbf{SR} \mathbf{R}^\top \mathbf{STC}^\top && \text{(second product)} \\
&= \mathbf{R}^\top \mathbf{S} (\mathbf{RR}^\top) \mathbf{STC}^\top && \text{(Theorem 5 in Matrices)} \\
&= \mathbf{R}^\top \mathbf{I}_r \mathbf{STC}^\top && \text{(Definition 5 in Special Matrices and Decompositions)} \\
&= \mathbf{G}. && \text{(Theorem 7 in Matrices)}
\end{aligned}$$

□

**Example 0.17** (The Moore-Penrose inverse of a rank-one matrix). For  $\mathbf{A}$  in [Example 22 in Subspaces and Rank](#), the rank factorization  $\mathbf{C} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ ,  $\mathbf{R} = \begin{bmatrix} 1 & -2 & -2 \end{bmatrix}$  ([Example 40 in Subspaces and Rank](#)) gives

$$\begin{aligned}
\mathbf{C}^\top \mathbf{C} &= 1 + 9 \\
&= 10
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{RR}^\top &= 1 + 4 + 4 \\
&= 9,
\end{aligned}$$

so

$$\begin{aligned}\mathbf{A}^+ &= \frac{1}{9 \cdot 10} \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \begin{bmatrix} 1 & 3 \end{bmatrix} \\ &= \frac{1}{90} \begin{bmatrix} 1 & 3 \\ -2 & -6 \\ -2 & -6 \end{bmatrix}.\end{aligned}$$

As a check on condition 1,

$$\begin{aligned}\mathbf{A}\mathbf{A}^+ &= \mathbf{C}\mathbf{T}\mathbf{C}^\top \\ &= \frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix},\end{aligned}$$

and its first row times  $\mathbf{A}$  is  $\frac{1}{10}(1 \cdot (1, -2, -2) + 3 \cdot (3, -6, -6)) = (1, -2, -2)$ , the first row of  $\mathbf{A}$ ; the second row is 3 times the first, like the second row of  $\mathbf{A}$ .

**Definition 0.6** (Consistent linear system). A linear system ([Definition 12 in Matrices](#))  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ , with  $\mathbf{A}$  an  $m \times n$  matrix and  $\tilde{\mathbf{b}} \in \mathbb{R}^m$ , is **consistent** if it has at least one solution  $\tilde{\mathbf{x}} \in \mathbb{R}^n$ , and **inconsistent** otherwise.

**Example 0.18** (A consistent system and an inconsistent one). With  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ ,  $\mathbf{A}(x_1, x_2) = (x_1, 0)$  ([Definition 11 in Matrices](#)).

- $\mathbf{A}\tilde{\mathbf{x}} = (2, 0)$  is consistent:  $\tilde{\mathbf{x}} = (2, 0)$  is a solution.
- $\mathbf{A}\tilde{\mathbf{x}} = (2, 1)$  is inconsistent: the second entry of  $\mathbf{A}\tilde{\mathbf{x}}$  is always 0, never 1.

**Theorem 0.13** (When a linear system has a solution). Let  $\mathbf{A}$  be  $m \times n$ , let  $\tilde{\mathbf{b}} \in \mathbb{R}^m$ , and let  $\mathbf{G}$  be any generalized inverse of  $\mathbf{A}$  ([Definition 0.4](#)). The following statements are equivalent:

1.  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  is consistent ([Definition 0.6](#));
2.  $\tilde{\mathbf{b}} \in \mathcal{C}(\mathbf{A})$  ([Definition 9 in Subspaces and Rank](#));
3.  $\mathbf{A}\mathbf{G}\tilde{\mathbf{b}} = \tilde{\mathbf{b}}$ .

### **i** Proof

*Proof.* **1 and 2 are equivalent.** Statement 1 says  $\tilde{\mathbf{b}} = \mathbf{A}\tilde{\mathbf{x}}$  for some  $\tilde{\mathbf{x}}$  ([Definition 0.6](#)), and  $\mathcal{C}(\mathbf{A})$  is the set of vectors  $\mathbf{A}\tilde{\mathbf{x}}$  ([Definition 9 in Subspaces and Rank](#)), so  $\tilde{\mathbf{b}}$  is in it exactly when  $\tilde{\mathbf{b}} = \mathbf{A}\tilde{\mathbf{x}}$  for some  $\tilde{\mathbf{x}}$ .

**1 implies 3.** By Theorem 0.9,  $\mathbf{G}\tilde{\mathbf{b}}$  is a solution, so  $\mathbf{A}(\mathbf{G}\tilde{\mathbf{b}}) = \tilde{\mathbf{b}}$ , which is statement 3 (Theorem 5 in Matrices).

**3 implies 1.** If  $\mathbf{A}\mathbf{G}\tilde{\mathbf{b}} = \tilde{\mathbf{b}}$ , then  $\tilde{\mathbf{x}} = \mathbf{G}\tilde{\mathbf{b}}$  is a solution, since  $\mathbf{A}(\mathbf{G}\tilde{\mathbf{b}}) = (\mathbf{A}\mathbf{G})\tilde{\mathbf{b}}$  (Theorem 5 in Matrices).  $\square$

**Example 0.19** (Testing consistency with a generalized inverse). For  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$  and the generalized inverse  $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$ ,  $\mathbf{A}\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 0 & 0 \end{bmatrix}$ .

- $\tilde{\mathbf{b}} = (2, 0)$ :

$$\begin{aligned} \mathbf{A}\mathbf{G}\tilde{\mathbf{b}} &= (2, 0) \\ &= \tilde{\mathbf{b}}, \end{aligned}$$

so the system is consistent.

- $\tilde{\mathbf{b}} = (2, 1)$ :  $\mathbf{A}\mathbf{G}\tilde{\mathbf{b}} = (7, 0) \neq \tilde{\mathbf{b}}$ , so it is not, as Example 0.13 found directly.

**Theorem 0.14** (All solutions of a linear system). *Let  $\mathbf{A}$  be  $m \times n$  with a generalized inverse  $\mathbf{G}$  (Definition 0.4).*

1.  $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{I}_n - \mathbf{GA})$ :  $\tilde{\mathbf{x}}$  solves  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{0}}_m$  exactly when  $\tilde{\mathbf{x}} = (\mathbf{I}_n - \mathbf{GA})\tilde{\mathbf{q}}$  for some  $\tilde{\mathbf{q}} \in \mathbb{R}^n$ .
2. If  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  is consistent (Definition 0.6), then  $\tilde{\mathbf{x}}$  is a solution exactly when

$$\tilde{\mathbf{x}} = \mathbf{G}\tilde{\mathbf{b}} + (\mathbf{I}_n - \mathbf{GA})\tilde{\mathbf{q}} \quad \text{for some } \tilde{\mathbf{q}} \in \mathbb{R}^n :$$

one particular solution plus a vector of  $\mathcal{N}(\mathbf{A})$ .

## **i** Proof

*Proof.* **Part 1.** First,

$$\begin{aligned} \mathbf{A}(\mathbf{I}_n - \mathbf{GA}) &= \mathbf{A}\mathbf{I}_n - \mathbf{A}(\mathbf{GA}) && \text{(Theorem 0.7)} \\ &= \mathbf{A}\mathbf{I}_n - \mathbf{AGA} && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{A} - \mathbf{AGA} && \text{(Theorem 7 in Matrices)} \\ &= \mathbf{A} - \mathbf{A} && \text{(Definition 0.4)} \\ &= \mathbf{0}_{m \times n}, && \text{(arithmetic)} \end{aligned}$$

so  $\mathbf{A}(\mathbf{I}_n - \mathbf{GA})\tilde{q} = \tilde{0}_m$  for every  $\tilde{q}$  (Theorem 5 in Matrices), and  $\mathcal{C}(\mathbf{I}_n - \mathbf{GA}) \subseteq \mathcal{N}(\mathbf{A})$ . Conversely, if  $\mathbf{A}\tilde{x} = \tilde{0}_m$ , then

$$\begin{aligned} (\mathbf{I}_n - \mathbf{GA})\tilde{x} &= \mathbf{I}_n\tilde{x} - (\mathbf{GA})\tilde{x} && \text{(Theorem 0.7, with } \tilde{x} \text{ as the last factor)} \\ &= \tilde{x} - (\mathbf{GA})\tilde{x} && \text{(Theorem 7 in Matrices)} \\ &= \tilde{x} - \mathbf{G}(\mathbf{A}\tilde{x}) && \text{(Theorem 5 in Matrices)} \\ &= \tilde{x} - \mathbf{G}\tilde{0}_m && (\mathbf{A}\tilde{x} = \tilde{0}_m) \\ &= \tilde{x}, && (\mathbf{G}\tilde{0}_m = \tilde{0}_n) \end{aligned}$$

so  $\tilde{x}$  is in  $\mathcal{C}(\mathbf{I}_n - \mathbf{GA})$ , with  $\tilde{q} = \tilde{x}$ .

**Part 2.**  $\mathbf{G}\tilde{b}$  is a solution (Theorem 0.9). For any  $\tilde{x}$ ,

$$\begin{aligned} \mathbf{A}(\tilde{x} - \mathbf{G}\tilde{b}) &= \mathbf{A}\tilde{x} - \mathbf{A}(\mathbf{G}\tilde{b}) && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \mathbf{A}\tilde{x} - \mathbf{AG}\tilde{b} && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{A}\tilde{x} - \tilde{b}, && \text{(Theorem 0.13, statement 3)} \end{aligned}$$

so  $\tilde{x}$  solves  $\mathbf{A}\tilde{x} = \tilde{b}$  exactly when  $\tilde{x} - \mathbf{G}\tilde{b} \in \mathcal{N}(\mathbf{A})$ , that is, by part 1, exactly when  $\tilde{x} - \mathbf{G}\tilde{b} = (\mathbf{I}_n - \mathbf{GA})\tilde{q}$  for some  $\tilde{q}$ .  $\square$

**Example 0.20** (All solutions of a consistent system). For  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ ,  $\mathbf{G} = \begin{bmatrix} 1 & 5 \\ 7 & 0 \end{bmatrix}$  and  $\tilde{b} = (2, 0)$ :  $\mathbf{G}\tilde{b} = (2, 14)$ ,  $\mathbf{GA} = \begin{bmatrix} 1 & 0 \\ 7 & 0 \end{bmatrix}$ , and  $\mathbf{I}_2 - \mathbf{GA} = \begin{bmatrix} 0 & 0 \\ -7 & 1 \end{bmatrix}$ , so

$$\tilde{x} = (2, 14) + (0, -7q_1 + q_2), \quad q_1, q_2 \in \mathbb{R}.$$

Since  $-7q_1 + q_2$  can be any number, the solutions are exactly the vectors  $(2, t)$ ,  $t \in \mathbb{R}$ , which is what

$$\begin{aligned} \mathbf{A}(x_1, x_2) &= (x_1, 0) \\ &= (2, 0) \end{aligned}$$

says directly.

**Corollary 0.1** (Solvable for every right-hand side, and solvable uniquely). *Let  $\mathbf{A}$  be  $m \times n$ .*

1.  $\mathbf{A}\tilde{x} = \tilde{b}$  is consistent for every  $\tilde{b} \in \mathbb{R}^m$  exactly when  $\text{rank}(\mathbf{A}) = m$ .
2. A consistent system  $\mathbf{A}\tilde{x} = \tilde{b}$  has exactly one solution exactly when  $\text{rank}(\mathbf{A}) = n$ .

**i** Proof

*Proof. Part 1.* By Theorem 0.13, the system is consistent for every  $\tilde{b}$  exactly when  $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$ .  $\mathcal{C}(\mathbf{A})$  is a subspace of  $\mathbb{R}^m$  (Theorem 12 in Subspaces and Rank) with dimension  $\text{rank}(\mathbf{A})$  (Theorem 16 in Subspaces and Rank), and  $\dim(\mathbb{R}^m) = m$  (Example 16 in Subspaces and Rank). If the dimension is  $m$ , then  $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$  (Theorem 11 in Subspaces and Rank); if  $\mathcal{C}(\mathbf{A}) = \mathbb{R}^m$ , the dimension is  $m$ .

**Part 2.** By Theorem 0.14, the solutions are a particular solution plus the vectors of  $\mathcal{N}(\mathbf{A})$ , so there is exactly one solution exactly when  $\mathcal{N}(\mathbf{A}) = \{\tilde{0}_n\}$ , that is, when  $\text{nullity}(\mathbf{A}) = 0$  (a subspace of dimension 0 has the empty list as a basis, whose span is  $\{\tilde{0}\}$ , by Definition 8 in Subspaces and Rank and Definition 5 in Subspaces and Rank; and  $\{\tilde{0}\}$  has dimension 0, by Example 16 in Subspaces and Rank). By Theorem 17 in Subspaces and Rank, that is when  $\text{rank}(\mathbf{A}) = n$ .  $\square$

**Example 0.21** (Three systems).

- The  $2 \times 3$  matrix  $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$  of Example 34 in Subspaces and Rank has rank  $2 = m$ , so every system with it is consistent, but  $2 < 3 = n$ , so the solutions are never unique.
- The  $3 \times 2$  matrix  $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$  of Example 0.1 has rank  $2 = n$ , so a consistent system has one solution, but  $2 < 3 = m$ , so some systems are inconsistent:  $(1, 0, 0) \notin \mathcal{C}(\mathbf{X})$ , because  $a(1, 1, 1) + b(1, 2, 3)$  has equal differences between consecutive entries, and  $(1, 0, 0)$  does not.
- An invertible  $p \times p$  matrix  $\mathbf{A}$  has rank  $p$ : if  $\mathbf{A}\tilde{x} = \tilde{0}$  then

$$\begin{aligned} \tilde{x} &= \mathbf{A}^{-1}\mathbf{A}\tilde{x} \\ &= \tilde{0}, \end{aligned}$$

so its nullity is 0 and its rank is  $p$  (Theorem 17 in Subspaces and Rank). So every system with it is consistent (part 1, with  $p = m$ ) and has exactly one solution (part 2, with  $p = n$ ), namely  $\mathbf{A}^{-1}\tilde{b}$ , since  $\mathbf{A}(\mathbf{A}^{-1}\tilde{b}) = \tilde{b}$ .

**Theorem 0.15** (The projection onto a column space, with any generalized inverse). *Let  $\mathbf{X}$  be an  $n \times p$  matrix of any rank, and let  $\mathbf{G}$  be any generalized inverse of  $\mathbf{X}^\top \mathbf{X}$  (Definition 0.4). Then for every  $\tilde{y} \in \mathbb{R}^n$ ,  $\mathbf{XGX}^\top \tilde{y}$  is the orthogonal projection of  $\tilde{y}$  onto*

$\mathcal{C}(\mathbf{X})$  (Definition 0.3). So the matrix  $\mathbf{X}\mathbf{G}\mathbf{X}^\top$  is the same for every choice of  $\mathbf{G}$ , and when  $\text{rank}(\mathbf{X}) = p$  it is the hat matrix  $\mathbf{H}$  (Definition 0.2).

### **i** Proof

*Proof.*  $\mathbf{X}^\top \tilde{\mathbf{y}}$  is in  $\mathcal{C}(\mathbf{X}^\top \mathbf{X})$ .  $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ , since  $(\mathbf{X}^\top \mathbf{X}) \tilde{\mathbf{z}} = \mathbf{X}^\top (\mathbf{X} \tilde{\mathbf{z}})$  (Theorem 5 in Matrices). Both are subspaces of  $\mathbb{R}^p$  (Theorem 12 in Subspaces and Rank) with the same dimension:  $\text{rank}(\mathbf{X}^\top \mathbf{X}) = \text{rank}(\mathbf{X}^\top)$  (Theorem 19 in Subspaces and Rank, Theorem 16 in Subspaces and Rank). So  $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$  (Theorem 11 in Subspaces and Rank), and since  $\mathbf{X}^\top \tilde{\mathbf{y}} \in \mathcal{C}(\mathbf{X}^\top)$  (Definition 9 in Subspaces and Rank),  $\mathbf{X}^\top \tilde{\mathbf{y}} = \mathbf{X}^\top \mathbf{X} \tilde{\mathbf{z}}$  for some  $\tilde{\mathbf{z}} \in \mathbb{R}^p$ .

**The split.** Let

$$\begin{aligned} \tilde{\mathbf{u}} &= \mathbf{X}\mathbf{G}\mathbf{X}^\top \tilde{\mathbf{y}} \\ &= \mathbf{X}(\mathbf{G}\mathbf{X}^\top \tilde{\mathbf{y}}) \end{aligned}$$

(Theorem 5 in Matrices), which is in  $\mathcal{C}(\mathbf{X})$  (Definition 9 in Subspaces and Rank). For  $\tilde{\mathbf{y}} - \tilde{\mathbf{u}}$ ,

$$\begin{aligned} \mathbf{X}^\top (\tilde{\mathbf{y}} - \tilde{\mathbf{u}}) &= \mathbf{X}^\top \tilde{\mathbf{y}} - \mathbf{X}^\top \tilde{\mathbf{u}} && \text{(Theorem 4 in Subspaces and Rank, with coefficients 1 and } -1) \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - \mathbf{X}^\top (\mathbf{X}\mathbf{G}\mathbf{X}^\top \tilde{\mathbf{y}}) && \text{(substitute } \tilde{\mathbf{u}}) \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \tilde{\mathbf{y}}) && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) \tilde{\mathbf{z}} && \text{(first step)} \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - ((\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X})) \tilde{\mathbf{z}} && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - (\mathbf{X}^\top \mathbf{X}) \tilde{\mathbf{z}} && \text{(Definition 0.4)} \\ &= \mathbf{X}^\top \tilde{\mathbf{y}} - \mathbf{X}^\top \tilde{\mathbf{y}} && \text{(first step)} \\ &= \tilde{\mathbf{0}}_p, && \text{(arithmetic)} \end{aligned}$$

so  $\tilde{\mathbf{y}} - \tilde{\mathbf{u}} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$  (Theorem 30 in Subspaces and Rank). By the uniqueness in Theorem 31 in Subspaces and Rank,  $\tilde{\mathbf{u}}$  is the orthogonal projection of  $\tilde{\mathbf{y}}$  onto  $\mathcal{C}(\mathbf{X})$ .

**Same matrix.** Every choice of  $\mathbf{G}$  gives a matrix that projects every  $\tilde{\mathbf{y}}$  onto  $\mathcal{C}(\mathbf{X})$ , so all choices give the same matrix (Theorem 0.5, part 2). When  $\text{rank}(\mathbf{X}) = p$ ,  $(\mathbf{X}^\top \mathbf{X})^{-1}$  is one choice (Theorem 0.1, Theorem 0.8), and it gives  $\mathbf{H}$ .  $\square$

**Example 0.22** (A design matrix with a repeated column). Let  $\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ , an intercept column entered twice, so  $\text{rank}(\mathbf{X}) = 1 < 2$  and  $\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$  is not invertible: its rank

is  $\text{rank}(\mathbf{X}) = 1 < 2$  ([Theorem 19 in Subspaces and Rank](#)), while an invertible  $2 \times 2$  matrix has rank 2 ([Example 0.21](#)). The matrix  $\mathbf{G} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{bmatrix}$  is a generalized inverse of it:

$$\begin{aligned} (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) &= \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \quad (\text{Definition 7 in Matrices, for the first product}) \\ &= \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}. \quad (\text{Definition 7 in Matrices}) \end{aligned}$$

Then

$$\begin{aligned} \mathbf{X} \mathbf{G} \mathbf{X}^\top &= \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad (\text{Definition 7 in Matrices, for } \mathbf{X} \mathbf{G}) \\ &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}, \quad (\text{Definition 7 in Matrices}) \end{aligned}$$

the hat matrix of the intercept-only model ([Example 0.2](#)). Another generalized inverse,  $\mathbf{G}' = \begin{bmatrix} 0 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$ , which is one because  $(\mathbf{X}^\top \mathbf{X}) \mathbf{G}' = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$  and that times  $\mathbf{X}^\top \mathbf{X}$  is again  $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ , gives  $\mathbf{X} \mathbf{G}' = \begin{bmatrix} 0 & \frac{1}{2} \\ 0 & \frac{1}{2} \end{bmatrix}$  and the same  $\mathbf{X} \mathbf{G}' \mathbf{X}^\top = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ , as [Theorem 0.15](#) says it must. Both project onto  $\mathcal{C}(\mathbf{X}) = \text{span}\{(1, 1)\}$ , the column space of the intercept-only design.

## Solving linear systems and least squares

### Source

This section is adapted from [Zhou \(2024b\)](#) and [Zhou \(2024a\)](#), used under the MIT License (see the license text in [Section 2 in Subspaces and Rank](#)). The first source states that every invertible matrix has an LU factorization; that is false without reordering rows (the  $2 \times 2$  row swap  $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  has none), so this section states only how an LU factorization, when one exists, solves a system. The least squares results are proved here from the closest-point theorem ([Theorem 0.3](#)) rather than by differentiating, and the QR factorization is derived from Gram-Schmidt ([Theorem 9 in Inner Products and Orthogonality](#)).

**Definition 0.7** (Triangular matrix). A square matrix  $\mathbf{U}$  is **upper triangular** if every entry below the diagonal is 0:  $u_{ij} = 0$  whenever  $i > j$ . A square matrix  $\mathbf{L}$  is **lower triangular** if every entry above the diagonal is 0:  $\ell_{ij} = 0$  whenever  $i < j$ . A lower

triangular matrix whose diagonal entries are all 1 is **unit lower triangular**.

**Example 0.23** (Triangular and not).

- $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$  is upper triangular: its entries below the diagonal, in positions  $(2, 1)$ ,  $(3, 1)$  and  $(3, 2)$ , are 0.
- $\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}$  is unit lower triangular.
- A diagonal matrix ([Definition 4 in Special Matrices and Decompositions](#)) is both upper and lower triangular.
- $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  is neither: its  $(2, 1)$  entry is not 0, and neither is its  $(1, 2)$  entry.

**Theorem 0.16** (Solving a triangular system by substitution). *Let  $\mathbf{U}$  be an  $n \times n$  upper triangular matrix ([Definition 0.7](#)) whose diagonal entries  $u_{11}, \dots, u_{nn}$  are all nonzero. For every  $\tilde{\mathbf{b}} \in \mathbb{R}^n$ , the system  $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  has exactly one solution, given for  $i = n, n-1, \dots, 1$  in turn by*

$$x_i = \frac{1}{u_{ii}} \left( b_i - \sum_{j=i+1}^n u_{ij} x_j \right)$$

*(back substitution). Likewise, if  $\mathbf{L}$  is lower triangular with nonzero diagonal,  $\mathbf{L}\tilde{\mathbf{y}} = \tilde{\mathbf{b}}$  has exactly one solution, given for  $i = 1, \dots, n$  by  $y_i = \frac{1}{\ell_{ii}} \left( b_i - \sum_{j=1}^{i-1} \ell_{ij} y_j \right)$  (**forward substitution**).*

### Proof

*Proof.* Row  $i$  of  $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  reads

$$\begin{aligned} b_i &= \sum_{j=1}^n u_{ij} x_j && \text{(Definition 11 in Matrices)} \\ &= \sum_{j=i}^n u_{ij} x_j && (u_{ij} = 0 \text{ for } j < i) \\ &= u_{ii} x_i + \sum_{j=i+1}^n u_{ij} x_j. && \text{(split off the } j = i \text{ term)} \end{aligned}$$

This equation is equivalent to each of

$$u_{ii} x_i = b_i - \sum_{j=i+1}^n u_{ij} x_j \quad (\text{subtract the sum from both sides})$$

$$x_i = \frac{1}{u_{ii}} \left( b_i - \sum_{j=i+1}^n u_{ij} x_j \right). \quad (\text{divide by } u_{ii} \neq 0)$$

Row  $n$  involves only  $x_n$ , so it fixes  $x_n$ ; once  $x_{i+1}, \dots, x_n$  are fixed, row  $i$  fixes  $x_i$ . So the rows, taken from the last to the first, hold exactly when  $\tilde{x}$  is the vector the formula builds: there is one solution, and only one. For  $\mathbf{L}$ , row  $i$  reads  $\sum_{j=1}^{i-1} \ell_{ij} y_j + \ell_{ii} y_i = b_i$ , and the same argument runs from the first row to the last.  $\square$

**Example 0.24** (Back substitution on a  $3 \times 3$  system). Solve  $\begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix} \tilde{x} = \begin{bmatrix} 8 \\ 1 \\ 1 \end{bmatrix}$ :

1.

$$\begin{aligned} x_3 &= \frac{1}{-1} \cdot 1 \\ &= -1; \end{aligned}$$

2.

$$\begin{aligned} x_2 &= \frac{1}{1/2} \left( 1 - \frac{1}{2} \cdot (-1) \right) \\ &= 2 \cdot \frac{3}{2} \\ &= 3; \end{aligned}$$

3.

$$\begin{aligned} x_1 &= \frac{1}{2} (8 - 1 \cdot 3 - (-1)(-1)) \\ &= \frac{1}{2} \cdot 4 \\ &= 2. \end{aligned}$$

So  $\tilde{x} = (2, 3, -1)$ . Without a nonzero diagonal the method fails:  $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \tilde{x} = (1, 1)$  has no solution, since its second row reads  $0 = 1$ .

**Theorem 0.17** (QR factorization). *Let  $\mathbf{A}$  be an  $m \times n$  matrix with  $\text{rank}(\mathbf{A}) = n$ . Then*

$$\underbrace{\mathbf{A}}_{m \times n} = \underbrace{\mathbf{Q}}_{m \times n} \underbrace{\mathbf{R}}_{n \times n},$$

where the columns of  $\mathbf{Q}$  are orthonormal ([Definition 17 in Vectors](#)) and  $\mathbf{R}$  is upper

triangular (Definition 0.7) with positive diagonal entries.

### **i** Proof

*Proof.* The  $n$  columns  $\tilde{a}_1, \dots, \tilde{a}_n$  of  $\mathbf{A}$  are linearly independent (Definition 3 in Subspaces and Rank), so the Gram-Schmidt process (Definition 10 in Inner Products and Orthogonality) runs all  $n$  steps and gives orthonormal  $\tilde{q}_1, \dots, \tilde{q}_n$  (Theorem 9 in Inner Products and Orthogonality, parts 1 and 3); let  $\mathbf{Q}$  have these columns. At step  $i$ ,  $\tilde{q}_i \neq \tilde{0}$ , and rearranging the orthogonalize step,

$$\begin{aligned}\tilde{a}_i &= \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j && \text{(add the sum to both sides of the orthogonalize step)} \\ &= \|\tilde{q}_i\| \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j. && \text{(normalize step: } \tilde{q}_i = \|\tilde{q}_i\| \tilde{q}_i \text{)}\end{aligned}$$

Let  $\mathbf{R}$  be the  $n \times n$  matrix with  $r_{ji} = \tilde{q}_j \cdot \tilde{a}_i$  for  $j < i$ ,  $r_{ii} = \|\tilde{q}_i\| > 0$ , and  $r_{ji} = 0$  for  $j > i$ ; it is upper triangular with positive diagonal. Column  $i$  of  $\mathbf{QR}$  is  $\mathbf{Q}$  times column  $i$  of  $\mathbf{R}$  (Definition 7 in Matrices), and

$$\begin{aligned}\mathbf{Q}(r_{1i}, \dots, r_{ni}) &= \sum_{j=1}^n r_{ji} \tilde{q}_j && \text{(Theorem 3 in Subspaces and Rank)} \\ &= \sum_{j=1}^i r_{ji} \tilde{q}_j && (r_{ji} = 0 \text{ for } j > i) \\ &= r_{ii} \tilde{q}_i + \sum_{j=1}^{i-1} r_{ji} \tilde{q}_j && \text{(split off the } j = i \text{ term)} \\ &= \|\tilde{q}_i\| \tilde{q}_i + \sum_{j=1}^{i-1} (\tilde{q}_j \cdot \tilde{a}_i) \tilde{q}_j && \text{(the entries of } \mathbf{R} \text{)} \\ &= \tilde{a}_i. && \text{(the display)}\end{aligned}$$

So  $\mathbf{QR} = \mathbf{A}$ . □

**Example 0.25** (A QR factorization from Gram-Schmidt). For  $\mathbf{A}$  with columns  $\tilde{a}_1 = (1, 1, 0)$ ,  $\tilde{a}_2 = (1, 0, 1)$ ,  $\tilde{a}_3 = (0, 1, 1)$ , Example 21 in Inner Products and Orthogonality and Example 23 in Inner Products and Orthogonality found  $\tilde{q}_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$ ,  $\tilde{q}_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$ ,  $\tilde{q}_3 = \frac{1}{\sqrt{3}}(-1, 1, 1)$ , with  $\|\tilde{q}_1\| = \sqrt{2}$ ,  $\|\tilde{q}_2\| = \sqrt{3/2}$ ,  $\|\tilde{q}_3\| = 2/\sqrt{3}$ ,

$$\begin{aligned}\tilde{q}_1 \cdot \tilde{a}_2 &= \tilde{q}_1 \cdot \tilde{a}_3 \\ &= \frac{1}{\sqrt{2}}\end{aligned}$$

and  $\tilde{q}_2 \cdot \tilde{a}_3 = \frac{1}{\sqrt{6}}$ . So

$$\mathbf{R} = \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3/2} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}.$$

As a check on the second column,

$$\begin{aligned}\frac{1}{\sqrt{2}} \tilde{q}_1 + \sqrt{3/2} \tilde{q}_2 &= \frac{1}{2} (1, 1, 0) + \frac{1}{2} (1, -1, 2) \\ &= (1, 0, 1) \\ &= \tilde{a}_2.\end{aligned}$$

**Definition 0.8** (Least squares solution). Let  $\mathbf{A}$  be  $m \times n$  and  $\tilde{b} \in \mathbb{R}^m$ . A **least squares solution** of  $\mathbf{A}\tilde{x} = \tilde{b}$  is a vector  $\hat{x} \in \mathbb{R}^n$  such that

$$\|\tilde{b} - \mathbf{A}\hat{x}\| \leq \|\tilde{b} - \mathbf{A}\tilde{x}\| \quad \text{for every } \tilde{x} \in \mathbb{R}^n.$$

The equations  $\mathbf{A}^\top \mathbf{A}\tilde{x} = \mathbf{A}^\top \tilde{b}$  are the **normal equations**.

**Example 0.26** (Fitting a constant). Let  $\mathbf{A} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  and  $\tilde{b} = (1, 3)$ . The system  $x(1, 1) = (1, 3)$  is inconsistent (Definition 0.6): its first entry needs  $x = 1$  and its second needs  $x = 3$ . For any number  $x$ ,

$$\begin{aligned}\|\tilde{b} - \mathbf{A}x\|^2 &= (1 - x)^2 + (3 - x)^2 \\ &= 2(x - 2)^2 + 2,\end{aligned}$$

which is smallest at  $x = 2$ , so  $\hat{x} = 2$  is the least squares solution. The value  $x = 1$  is not one: it gives  $0 + 4 = 4 > 2$ . The normal equations read  $2x = 4$ , and their solution is  $x = 2$  too.

**Theorem 0.18** (Least squares solutions solve the normal equations). Let  $\mathbf{A}$  be  $m \times n$  and  $\tilde{b} \in \mathbb{R}^m$ .

1.  $\hat{x}$  is a least squares solution (Definition 0.8) exactly when it solves the normal equations  $\mathbf{A}^\top \mathbf{A}\hat{x} = \mathbf{A}^\top \tilde{b}$ .
2. The normal equations are consistent (Definition 0.6).

3. Every least squares solution gives the same fitted vector  $\mathbf{A}\hat{\tilde{x}}$ : the orthogonal projection of  $\tilde{b}$  onto  $\mathcal{C}(\mathbf{A})$  (Definition 0.3).
4. The least squares solution is unique exactly when  $\text{rank}(\mathbf{A}) = n$ , and then  $\hat{\tilde{x}} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \tilde{b}$ .

### **i** Proof

*Proof.* Let  $\tilde{u}$  be the orthogonal projection of  $\tilde{b}$  onto  $\mathcal{C}(\mathbf{A})$ . The vectors  $\mathbf{A}\tilde{x}$ , as  $\tilde{x}$  ranges over  $\mathbb{R}^n$ , are exactly the points of  $\mathcal{C}(\mathbf{A})$  (Definition 9 in Subspaces and Rank). **Least squares solutions are the solutions of  $\mathbf{A}\tilde{x} = \tilde{u}$ .** If  $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$ , then for every  $\tilde{x}$ ,  $\mathbf{A}\tilde{x} \in \mathcal{C}(\mathbf{A})$ , so  $\|\tilde{b} - \mathbf{A}\hat{\tilde{x}}\| = \|\tilde{b} - \tilde{u}\| \leq \|\tilde{b} - \mathbf{A}\tilde{x}\|$  (Theorem 0.3), and  $\hat{\tilde{x}}$  is a least squares solution. Conversely, let  $\hat{\tilde{x}}$  be a least squares solution.  $\tilde{u} \in \mathcal{C}(\mathbf{A})$ , so  $\tilde{u} = \mathbf{A}\tilde{x}_0$  for some  $\tilde{x}_0$  (Definition 9 in Subspaces and Rank), and

$$\begin{aligned} \|\tilde{b} - \mathbf{A}\hat{\tilde{x}}\| &\leq \|\tilde{b} - \mathbf{A}\tilde{x}_0\| \quad (\text{Definition 0.8}) \\ &= \|\tilde{b} - \tilde{u}\| \quad (\tilde{u} = \mathbf{A}\tilde{x}_0) \\ &\leq \|\tilde{b} - \mathbf{A}\hat{\tilde{x}}\|, \quad (\text{Theorem 0.3, with } \tilde{w} = \mathbf{A}\hat{\tilde{x}} \in \mathcal{C}(\mathbf{A})) \end{aligned}$$

so the last inequality is an equality, and the equality case of Theorem 0.3 gives  $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$ . **Parts 1 and 3.**  $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$  holds exactly when  $\tilde{b} - \mathbf{A}\hat{\tilde{x}} \in \mathcal{C}(\mathbf{A})^\perp$ : if  $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$ , then  $\tilde{b} - \tilde{u} \in \mathcal{C}(\mathbf{A})^\perp$  (Definition 0.3); conversely, if  $\tilde{b} - \mathbf{A}\hat{\tilde{x}} \in \mathcal{C}(\mathbf{A})^\perp$ , then  $\tilde{b} = \mathbf{A}\hat{\tilde{x}} + (\tilde{b} - \mathbf{A}\hat{\tilde{x}})$  is a split into  $\mathcal{C}(\mathbf{A})$  and  $\mathcal{C}(\mathbf{A})^\perp$ , so  $\mathbf{A}\hat{\tilde{x}} = \tilde{u}$  by the uniqueness in Theorem 31 in Subspaces and Rank. And  $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$  (Theorem 30 in Subspaces and Rank), where

$$\begin{aligned} \mathbf{A}^\top (\tilde{b} - \mathbf{A}\hat{\tilde{x}}) &= \mathbf{A}^\top \tilde{b} - \mathbf{A}^\top (\mathbf{A}\hat{\tilde{x}}) \quad (\text{Theorem 0.7}) \\ &= \mathbf{A}^\top \tilde{b} - \mathbf{A}^\top \mathbf{A}\hat{\tilde{x}}, \quad (\text{Theorem 5 in Matrices}) \end{aligned}$$

which is  $\tilde{0}_n$  exactly when the normal equations hold.

**Part 2.** The vector  $\tilde{x}_0$  above satisfies  $\mathbf{A}\tilde{x}_0 = \tilde{u}$ , so it is a least squares solution, and it solves the normal equations by part 1.

**Part 4.** The least squares solutions are exactly the solutions of  $\mathbf{A}\tilde{x} = \tilde{u}$ , a consistent system, so there is exactly one when  $\text{rank}(\mathbf{A}) = n$  and more than one otherwise (Corollary 0.1). When  $\text{rank}(\mathbf{A}) = n$ ,  $\mathbf{A}^\top \mathbf{A}$  is invertible (Theorem 0.1), and with  $\mathbf{M} \stackrel{\text{def}}{=} \mathbf{A}^\top \mathbf{A}$ ,

$$\begin{aligned} \hat{\tilde{x}} &= \mathbf{I}_n \hat{\tilde{x}} \quad (\text{Theorem 7 in Matrices}) \\ &= (\mathbf{M}^{-1} \mathbf{M}) \hat{\tilde{x}} \quad (\text{Definition 5 in Special Matrices and Decompositions}) \\ &= \mathbf{M}^{-1} (\mathbf{M}\hat{\tilde{x}}) \quad (\text{Theorem 5 in Matrices}) \\ &= \mathbf{M}^{-1} \mathbf{A}^\top \tilde{b}. \quad (\text{the normal equations}) \end{aligned}$$

□

**Example 0.27** (A least squares line). Fit  $y = x_1 + x_2 t$  to the points  $(t, y) = (1, 1), (2, 2), (3, 2)$ :  $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$  (the  $\mathbf{X}$  of Example 0.1) and  $\tilde{\mathbf{b}} = (1, 2, 2)$ .  $\mathbf{A}$  has rank 2 (Example 2 in Subspaces and Rank), so the solution is unique (part 4).  $\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$  has inverse  $\frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$  (Example 0.1), and

$$\begin{aligned} \mathbf{A}^\top \tilde{\mathbf{b}} &= (1 + 2 + 2, 1 + 4 + 6) \\ &= (5, 11), \end{aligned}$$

so

$$\begin{aligned} \hat{\mathbf{x}} &= \frac{1}{6} (14 \cdot 5 - 6 \cdot 11, -6 \cdot 5 + 3 \cdot 11) \\ &= \frac{1}{6} (4, 3) \\ &= \left(\frac{2}{3}, \frac{1}{2}\right). \end{aligned}$$

The fitted values are  $\mathbf{A}\hat{\mathbf{x}} = (\frac{7}{6}, \frac{5}{3}, \frac{13}{6})$ , and the residuals  $\tilde{\mathbf{b}} - \mathbf{A}\hat{\mathbf{x}} = (-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6})$  are orthogonal to both columns of  $\mathbf{A}$ :  $-\frac{1}{6} + \frac{1}{3} - \frac{1}{6} = 0$  and  $-\frac{1}{6} + \frac{2}{3} - \frac{1}{2} = 0$ .

**Example 0.28** (Many least squares solutions, one fitted vector). With the rank-1 matrix

$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{bmatrix}$  (Example 2 in Subspaces and Rank) and  $\tilde{\mathbf{b}} = (1, 2, 2)$ ,  $\mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 12 \end{bmatrix}$  and

$\mathbf{A}^\top \tilde{\mathbf{b}} = (5, 10)$ , so both normal equations say  $3x_1 + 6x_2 = 5$ , that is,  $x_1 + 2x_2 = \frac{5}{3}$ . There are infinitely many least squares solutions, such as  $(\frac{5}{3}, 0)$  and  $(0, \frac{5}{6})$ , as part 4 predicts for  $\text{rank}(\mathbf{A}) = 1 < 2$ ; but every one gives the same fitted vector

$$\begin{aligned} \mathbf{A}\hat{\mathbf{x}} &= (x_1 + 2x_2) (1, 1, 1) \\ &= \frac{5}{3} (1, 1, 1), \end{aligned}$$

as part 3 says.

**Theorem 0.19** (Solving least squares by QR). *Let  $\mathbf{A}$  be  $m \times n$  with  $\text{rank}(\mathbf{A}) = n$ , with QR factorization  $\mathbf{A} = \mathbf{Q}\mathbf{R}$  (Theorem 0.17), and let  $\tilde{\mathbf{b}} \in \mathbb{R}^m$ . The least squares solution (Definition 0.8) is the unique solution of*

$$\underset{n \times n}{\mathbf{R}} \underset{n \times m}{\hat{\tilde{x}}} = \underset{n \times m}{\mathbf{Q}^\top} \tilde{b},$$

which back substitution computes (Theorem 0.16). When  $m = n$ , it is the unique solution of  $\mathbf{A}\tilde{x} = \tilde{b}$ .

### **i** Proof

*Proof.*  $\mathbf{R}$  is upper triangular with positive diagonal, so  $\mathbf{R}\tilde{x} = \mathbf{Q}^\top \tilde{b}$  has exactly one solution  $\hat{\tilde{x}}$  (Theorem 0.16).  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n$ , as in the proof of Theorem 0.4. Then

$$\begin{aligned} \mathbf{A}^\top \mathbf{A} \hat{\tilde{x}} &= (\mathbf{QR})^\top (\mathbf{QR}) \hat{\tilde{x}} && \text{(substitute } \mathbf{A} = \mathbf{QR}) \\ &= \mathbf{R}^\top \mathbf{Q}^\top (\mathbf{QR}) \hat{\tilde{x}} && \text{(Theorem 16 in Matrices)} \\ &= \mathbf{R}^\top (\mathbf{Q}^\top \mathbf{Q}) (\mathbf{R} \hat{\tilde{x}}) && \text{(Theorem 5 in Matrices)} \\ &= \mathbf{R}^\top \mathbf{I}_n (\mathbf{R} \hat{\tilde{x}}) && (\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_n) \\ &= \mathbf{R}^\top (\mathbf{R} \hat{\tilde{x}}) && \text{(Theorem 7 in Matrices)} \\ &= \mathbf{R}^\top (\mathbf{Q}^\top \tilde{b}) && (\mathbf{R} \hat{\tilde{x}} = \mathbf{Q}^\top \tilde{b}) \\ &= (\mathbf{R}^\top \mathbf{Q}^\top) \tilde{b} && \text{(Theorem 5 in Matrices)} \\ &= (\mathbf{QR})^\top \tilde{b} && \text{(Theorem 16 in Matrices)} \\ &= \mathbf{A}^\top \tilde{b}, && (\mathbf{A} = \mathbf{QR}) \end{aligned}$$

so  $\hat{\tilde{x}}$  solves the normal equations and is the least squares solution (Theorem 0.18), which is unique because  $\text{rank}(\mathbf{A}) = n$ . When  $m = n$ ,

$$\begin{aligned} \text{rank}(\mathbf{A}) &= n \\ &= m, \end{aligned}$$

so  $\mathbf{A}\tilde{x} = \tilde{b}$  is consistent for every  $\tilde{b}$  (Corollary 0.1, part 1), so its solution leaves residual  $\tilde{0}$ , is a least squares solution, and is therefore  $\hat{\tilde{x}}$ .  $\square$

**Example 0.29** (The least squares line by QR). For  $\mathbf{A}$  and  $\tilde{b}$  of Example 0.27, Gram-Schmidt gives  $\tilde{q}_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$ ; then

$$\begin{aligned} \tilde{q}_1 \cdot (1, 2, 3) &= \frac{6}{\sqrt{3}} \\ &= 2\sqrt{3}, \\ \tilde{q}_2 &= (1, 2, 3) - 2(1, 1, 1) \\ &= (-1, 0, 1) \end{aligned}$$

with norm  $\sqrt{2}$ , and  $\tilde{q}_2 = \frac{1}{\sqrt{2}}(-1, 0, 1)$ . So

$$\begin{aligned}\mathbf{R} &= \begin{bmatrix} \sqrt{3} & 2\sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}, \\ \mathbf{Q}^\top \tilde{\mathbf{b}} &= \left( \frac{1+2+2}{\sqrt{3}}, \frac{-1+0+2}{\sqrt{2}} \right) \\ &= \left( \frac{5}{\sqrt{3}}, \frac{1}{\sqrt{2}} \right).\end{aligned}$$

Back substitution gives

$$\begin{aligned}x_2 &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \\ &= \frac{1}{2}\end{aligned}$$

and

$$\begin{aligned}x_1 &= \frac{1}{\sqrt{3}} \left( \frac{5}{\sqrt{3}} - 2\sqrt{3} \cdot \frac{1}{2} \right) \\ &= \frac{5}{3} - 1 \\ &= \frac{2}{3},\end{aligned}$$

the solution found in Example 0.27.

**Definition 0.9** (LU factorization). An **LU factorization** of an  $n \times n$  matrix  $\mathbf{A}$  is a product  $\mathbf{A} = \mathbf{L}\mathbf{U}$  with  $\mathbf{L}$  unit lower triangular and  $\mathbf{U}$  upper triangular (Definition 0.7).

**Example 0.30** (An LU factorization of a  $3 \times 3$  matrix). Let  $\mathbf{A} = \begin{bmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{bmatrix}$ ,

$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{3}{2} & 1 & 0 \\ -1 & 4 & 1 \end{bmatrix}$  (unit lower triangular) and  $\mathbf{U} = \begin{bmatrix} 2 & 1 & -1 \\ 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & -1 \end{bmatrix}$  (upper triangular). By

Definition 7 in Matrices, entry  $(i, j)$  of  $\mathbf{L}\mathbf{U}$  is  $\sum_k \ell_{ik} u_{kj}$ , so row  $i$  of  $\mathbf{L}\mathbf{U}$  is  $\sum_k \ell_{ik}$  times row  $k$  of  $\mathbf{U}$ . Row 1 is  $1 \cdot (2, 1, -1) = (2, 1, -1)$ ; row 2 is  $-\frac{3}{2}(2, 1, -1) + (0, \frac{1}{2}, \frac{1}{2}) = (-3, -1, 2)$ , and row 3 is  $-(2, 1, -1) + 4(0, \frac{1}{2}, \frac{1}{2}) + (0, 0, -1) = (-2, 1, 2)$ . These are the rows of  $\mathbf{A}$ , so  $\mathbf{L}\mathbf{U} = \mathbf{A}$  is an LU factorization. The below-diagonal entries of  $\mathbf{L}$  are the negatives of the multiples of earlier rows that Gaussian elimination adds to reduce  $\mathbf{A}$  to  $\mathbf{U}$ .

**Example 0.31** (An invertible matrix with no LU factorization).  $\mathbf{P} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  is invertible:  $\mathbf{P}^2 = \mathbf{I}_2$ , so  $\mathbf{P}^{-1} = \mathbf{P}$  (Definition 5 in Special Matrices and Decompositions); but it has no LU factorization. If  $\mathbf{P} = \mathbf{L}\mathbf{U}$ , the  $(1, 1)$  entry gives  $0 = 1 \cdot u_{11}$ , so  $u_{11} = 0$ , and then

the  $(2, 1)$  entry gives

$$\begin{aligned} 1 &= \ell_{21} u_{11} \\ &= 0, \end{aligned}$$

which is impossible. Swapping the two rows first removes the obstacle: the swapped matrix is  $\mathbf{I}_2 = \mathbf{I}_2 \mathbf{I}_2$ .

**Theorem 0.20** (Solving a system with an LU factorization). *If  $\mathbf{A} = \mathbf{LU}$  is an LU factorization (Definition 0.9) and the diagonal entries of  $\mathbf{U}$  are all nonzero, then for every  $\tilde{\mathbf{b}} \in \mathbb{R}^n$  the system  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$  has exactly one solution: solve  $\mathbf{L}\tilde{\mathbf{y}} = \tilde{\mathbf{b}}$  by forward substitution, then  $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{y}}$  by back substitution (Theorem 0.16).*

### **i** Proof

*Proof.*  $\mathbf{L}$  has diagonal entries 1 and  $\mathbf{U}$  has nonzero ones, so each of the two triangular systems has exactly one solution (Theorem 0.16). If  $\tilde{\mathbf{y}}$  and  $\tilde{\mathbf{x}}$  are those solutions, then

$$\begin{aligned} \mathbf{A}\tilde{\mathbf{x}} &= \mathbf{L}(\mathbf{U}\tilde{\mathbf{x}}) \\ &= \mathbf{L}\tilde{\mathbf{y}} \\ &= \tilde{\mathbf{b}} \end{aligned}$$

(Theorem 5 in Matrices), so  $\tilde{\mathbf{x}}$  is a solution. Conversely, if  $\mathbf{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}$ , then  $\tilde{\mathbf{y}} \stackrel{\text{def}}{=} \mathbf{U}\tilde{\mathbf{x}}$  satisfies

$$\begin{aligned} \mathbf{L}\tilde{\mathbf{y}} &= \mathbf{L}(\mathbf{U}\tilde{\mathbf{x}}) \\ &= (\mathbf{LU})\tilde{\mathbf{x}} \\ &= \mathbf{A}\tilde{\mathbf{x}} \\ &= \tilde{\mathbf{b}} \end{aligned}$$

(Theorem 5 in Matrices), so  $\tilde{\mathbf{y}}$  is the unique solution of that system, and  $\tilde{\mathbf{x}}$  is then the unique solution of  $\mathbf{U}\tilde{\mathbf{x}} = \tilde{\mathbf{y}}$ .  $\square$

**Example 0.32** (Two triangular solves). With  $\mathbf{L}$  and  $\mathbf{U}$  of Example 0.30, solve  $\mathbf{A}\tilde{\mathbf{x}} = (8, -11, -3)$ . Forward substitution on  $\mathbf{L}\tilde{\mathbf{y}} = (8, -11, -3)$  gives  $y_1 = 8$ ,

$$\begin{aligned} y_2 &= -11 + \frac{3}{2} \cdot 8 \\ &= 1, \\ y_3 &= -3 + 8 - 4 \cdot 1 \\ &= 1. \end{aligned}$$

Back substitution on  $\mathbf{U}\tilde{x} = (8, 1, 1)$  is Example 0.24, which gives  $\tilde{x} = (2, 3, -1)$ . As a check,

$$\begin{aligned}\mathbf{A}(2, 3, -1) &= (4 + 3 + 1, -6 - 3 - 2, -4 + 3 - 2) \\ &= (8, -11, -3).\end{aligned}$$

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